CORE MATHEMATICS GRADE 10: PROBABILITY I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 3.0: Statistics and Probability
Sub-Strand	Sub-Strand 3.2: Probability I
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Experimental probability — tossing coins (Kenyan shillings), throwing dice; recording outcomes and computing relative frequency – Probability scale — identifying the range of probability from 0 (impossible) to 1 (certain) with real-life Kenyan examples – Probability space (sample space) — listing all possible outcomes of events systematically using tables and tree diagrams – Theoretical probability — calculating probability of a single event from the sample space – Mutually exclusive events — identifying events that cannot occur simultaneously; applying the Addition Law $P(A \cup B) = P(A) + P(B)$ – Independent events — identifying events where the outcome of one does not affect the other; applying the Multiplication Law $P(A \cap B) = P(A) \times P(B)$ – Laws of probability — addition law for mutually exclusive events and multiplication law for independent events – Probability tree diagrams — constructing and using tree diagrams to determine probabilities of two sequential independent and mutually exclusive events – Application of probability — applying probability concepts to real-life Kenyan situations (road safety, games, weather, market outcomes)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) perform experiments involving probabilities in different situations, b) identify the range of probability in different situations, c) generate probability space of different events, d) determine the probability of mutually exclusive and independent events, e) apply the laws of probability in different situations, f) determine the probability of independent events using tree diagrams, g) appreciate the application of probability in real-life situations.
Core Competencies	– Communication and Collaboration: the learner carries out experiments such as tossing Kenyan coins or throwing dice, records results with peers, discusses findings, and shares conclusions about experimental probability with the class. – Citizenship: the learner acknowledges Kenyan currency (10-shilling and 5-shilling coins) as they carry out tossing experiments, connecting mathematics to national identity and civic awareness. – Self-Efficacy: the learner appreciates his/her ability to design and play probability-based games, building confidence in

	applying mathematical reasoning to everyday Kenyan contexts such as predicting matatu arrival times or market outcomes at Wakulima Market.
Core Values	– Responsibility: the learner carefully tosses coins or throws dice to illustrate experimental probability, ensuring accurate recording of outcomes and honest reporting of results. – Social Justice: the learner plays probability games where they accept that there is winning or losing and acknowledges honest outcomes, recognising that fairness in games mirrors fairness in society. – Integrity: the learner records experimental outcomes accurately without manipulation, understanding that valid conclusions depend on truthful data collection.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners formulate questions about why experimental probability (e.g., from 50 coin tosses) differs from theoretical probability of 0.5, driving inquiry into the Law of Large Numbers in a Kenyan context. – Planning and Carrying Out Investigations: learners design and conduct structured probability experiments — tossing Kenyan shillings, rolling dice, drawing cards — specifying number of trials, recording tables, and identifying variables. – Analysing and Interpreting Data: learners tally frequency tables, compute relative frequencies across increasing trial counts, construct sample space tables, and read probability tree diagrams to draw evidence-based conclusions. – Using Mathematics and Computational Thinking: learners apply the addition law (mutually exclusive events) and multiplication law (independent events), compute probabilities as fractions, decimals, and percentages, and verify results against experimental data. – Obtaining, Evaluating, and Communicating Information: learners present probability findings — including road-safety pedestrian-crossing scenarios and game designs — to peers and evaluate the reasonableness of each other's probability claims.
Pertinent & Contemporary Issues (PCIs)	– Social Cohesion: the learner interacts freely with classmates as they discuss, design, and play probability-based games, building teamwork and respect across different groups within the Kenyan school community. – Life Skills and Values Education: the learner uses different experimental events and lists all possible outcomes, practising systematic decision-making skills applicable to real-life choices such as crop selection in Rift Valley farming households. – Road Safety: the learner discusses with peers the probability of pedestrians crossing roads at undesignated areas in Nairobi or Mombasa and highlights the statistical dangers for public awareness, linking mathematics to civic responsibility. – Financial Literacy: the learner connects probability to everyday economic decisions — such as the likelihood of a hawker at Gikomba Market selling out stock — reinforcing responsible financial reasoning.
Career Connections	– Actuarial Science / Insurance: probability is the mathematical foundation of risk assessment; Kenyan insurance companies such as Jubilee Insurance and AAR use probability to price premiums and model claims. – Data Science and Statistics: government agencies such as the Kenya National Bureau of Statistics (KNBS) and organisations like the Kenya Medical Research Institute (KEMRI) use probability models to design surveys and analyse health data. – Medicine and Public Health: epidemiologists at KEMRI and the Ministry of Health use probability to model disease outbreak likelihood, vaccine efficacy, and patient diagnosis uncertainty. – Agriculture and Meteorology: the Kenya Meteorological Department uses probability to issue rainfall and drought forecasts that guide planting decisions for maize and bean farmers in the Rift Valley and Western Kenya. – Engineering and Quality Control: Kenyan manufacturers and construction firms (e.g., on SGR infrastructure projects) use probability in quality-control sampling to determine the likelihood of defective materials. – Sports Analysis: probability underpins performance analytics for Kenyan athletes — from predicting marathon finishing times to assessing scoring probabilities in the Kenyan Premier League.
Focus for Lessons	This unit introduces Grade 10 learners to the foundational concepts of probability through a blend of hands-on experimentation, systematic reasoning, and real-life Kenyan contexts. Learners move from conducting simple coin-tossing and dice-rolling

	experiments to constructing sample spaces, applying the addition and multiplication laws, and drawing probability tree diagrams for sequential events. Throughout the 8-lesson sequence, a single anchoring phenomenon — the surprising unpredictability of a matatu conductor's daily passenger count at a Nairobi bus terminus — threads all concepts together, helping learners appreciate why probability is an indispensable tool for making informed decisions under uncertainty in Kenyan everyday life.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "How can Kamau — and other Kenyans facing uncertain everyday outcomes — use the mathematics of probability to make smarter, fairer, and safer decisions?" KICD KEY INQUIRY QUESTION: 1. How is probability applied in real-life situations?
Anchoring Phenomenon	ANCHORING PHENOMENON — "Why Can't the Matatu Conductor Predict Today's Full Bus?" At the busy Machakos Country Bus Station in Nairobi, a matatu conductor named Kamau notices something puzzling: even though his 14-seater matatu runs the Nairobi–Kitengela route every day at the same time, some mornings every seat is taken within 10 minutes, while other mornings only 6 passengers board in 30 minutes — with no obvious reason for the difference. Weather, day of the week, and school term all seem to matter sometimes but not always. Kamau keeps a tally notebook but still cannot reliably predict whether a given seat will be occupied on any single trip. Students are shown Kamau's tally notebook (a teacher-created table of 20 simulated trip records showing passenger counts of 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14) and asked: "Why is it impossible for Kamau to know for certain whether the next seat will be filled — yet we can still say something mathematical and useful about his chances?" This phenomenon is puzzling because learners initially expect patterns to be perfectly predictable, yet the data resists a simple rule. It anchors the entire unit: experimental probability explains why Kamau's tallies approximate — but never exactly match — theoretical probability; sample space shows all possible boarding outcomes; mutually exclusive events capture whether a seat is either filled or empty (not both); independent events explain why one passenger's decision does not force another passenger to board; tree diagrams let Kamau model two consecutive stops; and the laws of probability give him a rigorous tool to plan staffing and fuel budgets despite uncertainty.
Storyline Thread	Lesson 1 — Anchoring the Phenomenon & Experimental Probability: Learners are introduced to Kamau's matatu tally notebook. In pairs they analyse the 20-trip dataset, identify the unpredictability, and attempt to explain it. They then conduct their own experiment — tossing a 10-shilling coin 30 times each — recording heads/tails tallies and computing relative frequency. Class results are pooled on the board to observe how combined totals approach 0.5. Students add their first question to the Driving Question Board (DQB): "Why doesn't experimental probability exactly equal what we expect?" Lesson 2 — The Probability Scale (Range of Probability): Learners revisit Kamau's data and list events ranging from "impossible" (a matatu carries 20 passengers when it seats 14) to "certain" (the matatu will eventually stop). They classify a set of 12 Kenyan real-life events on a 0-to-1 number line (e.g., "it will rain in Kisumu during April," "a mandazi costs 0 shillings"). Students formalise the scale $0 \le P(E) \le 1$ and link it back to Kamau's notebook frequencies. DQB updated with: "Can we ever be 100% sure about Kamau's next trip?" Lesson 3 — Probability Space (Sample Space): Learners systematically list all possible outcomes for Kamau's simplified model: at each of two stops (Mlolongo and Athi River), a seat is either taken (T) or empty (E). They construct a two-way table and a list of the sample space {TT, TE, ET, EE}. They extend this to rolling two dice at a school game stall, drawing a 6×6 sample space grid.

	Theoretical probability is defined from the sample space. DQB updated: "How do we count all possibilities without missing any?" Lesson 4 — Theoretical Probability and Single Events: Using the sample space grids from Lesson 3, learners calculate theoretical probabilities of specific outcomes (e.g., both seats filled, at least one seat empty, sum of two dice = 7 at the game stall). They compare their computed values against the experimental frequencies from Lesson 1 and discuss convergence. Kamau's scenario is revisited: what is the theoretical probability that a randomly chosen seat on a given trip is occupied if historically 10 out of 14 seats are filled on average? DQB updated: "Are theoretical and experimental probabilities ever exactly the same?" Lesson 5 — Mutually Exclusive Events and the Addition Law: Learners explore the Kisumu school prize-draw scenario. They identify that winning Prize A and winning Prize B are mutually exclusive (cannot happen at the same time for the same student in a single draw). The Addition Law P(A or B) = P(A) + P(B) is derived through the sample space. Learners apply this to Kamau's scenario (probability that a trip is either fully booked OR has fewer than 7 passengers) and to coin/dice problems. DQB updated: "What makes two events mutually exclusive?" Lesson 6 — Independent Events and the Multiplication Law: Learners investigate the Kericho tea-farmer scenario and the coin-tossing experiment: does the result of the first toss affect the second? Through experimentation and sample space analysis, learners confirm independence and derive P(A and B) = P(A) × P(B). They contrast this with dependent events (e.g., drawing two passengers from a waiting queue without replacement — a preview of Probability II). Application: What is the probability that Kamau's matatu is full on both the morning and afternoon trip, if each trip is independent? DQB updated: "How do we know when two events are truly independent?" Lesson 7 — Probability Tree Diagrams: Learners construct tree diagrams for two sequential independent events in Kamau's context: Stop 1 (seat filled / empty) then Stop 2 (seat filled / empty). They label branches with probabilities and multiply along branches to find the probability of each outcome, then add across outcomes that satisfy a condition. They also model the school prize draw (mutually exclusive) on a tree diagram. A second tree models a road-safety scenario: the probability that a pedestrian crosses at an undesignated point in Nairobi AND is involved in a near-miss, raising civic awareness. DQB updated: "How do tree diagrams help us see all possible outcomes at once?" Lesson 8 — Application, Games, and Unit Synthesis: Learner groups design a simple probability-based game (e.g., a spinning wheel or card draw) for a school fete, specifying sample space, theoretical probabilities, and fairness conditions. Groups present their games and the class evaluates fairness using the laws from Lessons 5–7. The class returns to Kamau's full 20-trip dataset, computes the experimental probability of a fully booked trip, compares it with a theoretical model, and writes a brief "Probability Report" advising Kamau on staffing decisions. The DQB is reviewed: all questions are revisited and answered. Final cumulative model is completed: a annotated probability framework diagram connecting experimental probability → sample space → theoretical probability → addition and multiplication laws → tree diagrams → real-life decision making.
Supporting Phenomena	– Tossing a 10-shilling Kenyan coin 40 times: learners expect exactly 20 heads but consistently get a different number — raising the question of why experimental results differ from the theoretical 0.5, and how many trials are needed before the results stabilise (connects to experimental probability and the range of probability). – Rolling a single die at a githeri-cooking competition to decide who stirs the pot first: learners explore all possible outcomes (1–6) and realise some outcomes (rolling a 7) are impossible while others (rolling an odd number) are quite likely, grounding the probability scale from 0 to 1. – A Kericho tea farmer choosing between two seed varieties on back-to-back planting seasons: whether the first season's variety succeeds or fails does not change the probability of the second season's variety succeeding — a concrete Kenyan illustration of independent events and the multiplication law.

	– A school prize-giving draw at a Kisumu secondary school where two students cannot win the same prize simultaneously: this models mutually exclusive events and the addition law, making abstract algebra tangible through a familiar school ceremony.

LESSON 1 (80 minutes): Anchoring the Phenomenon & Experimental Probability: Why Can't Kamau Predict His Full Bus?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit phenomenon and establish that real-world outcomes (like Kamau's matatu passengers) are unpredictable in individual instances yet mathematically describable through experimental probability and relative frequency.
Knowledge	Learners will know that: (i) experimental probability is defined as the relative frequency of an event over repeated trials; (ii) experimental probability approaches — but does not always equal — the theoretical probability as the number of trials increases; (iii) a tally/frequency table is used to organise outcomes of repeated experiments.
Skills	Learners will be able to: (i) read and interpret a frequency tally table of real-world outcomes; (ii) conduct a coin-toss experiment (30 trials per pair) and record results accurately; (iii) calculate relative frequency (experimental probability) from tally data; (iv) pool class data and observe the convergence effect towards 0.5.
Attitudes	Learners will: (i) appreciate that uncertainty is a natural and mathematically manageable feature of everyday Kenyan life; (ii) show responsibility in recording experimental data honestly; (iii) demonstrate social justice values by accepting that chance outcomes — like winning or losing — must be acknowledged honestly.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	This is the opening lesson of the unit. Learners encounter Kamau's matatu tally notebook for the first time, surface their own misconceptions about predictability, and generate the central puzzle that will drive all subsequent lessons: why does repeated experiment approximate — but never perfectly match — a theoretical expectation? The coin-toss experiment provides learners' first personal dataset and anchors the vocabulary of experimental probability. The Driving Question Board is opened, seeding questions that later lessons will answer.
Safety Notes	No significant hazards. Remind learners to handle 10-shilling coins carefully and not to flip coins towards classmates' faces. Ensure coins are returned after the activity. Learners with motor difficulties may use a cup-shake method to toss the coin instead.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon — Kamau, a matatu conductor at Machakos Country Bus Station, cannot predict whether each seat on his 14-seater Nairobi–Kitengela matatu will be filled on any given trip, even after keeping 20 trips of careful tally records. In pairs, learners analyse Kamau's simulated dataset (passenger counts: 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14), attempt to find a perfect rule, and discover they cannot. Learners then conduct their own coin-toss experiment — tossing a 10-shilling coin 30 times each pair — recording heads and tails tallies and computing their own relative frequencies. Class results are pooled on the board to observe how the combined totals converge toward 0.5. The lesson closes with the opening of the Driving Question Board (DQB), on which learners post their first burning questions. This lesson covers the KICD learning experience: "carry out experiments such as tossing coins or throwing dice to illustrate experimental probability" and begins developing the core competencies of Communication and Collaboration and Citizenship (acknowledging Kenyan currency).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed or projected copy of Kamau's Tally Notebook — a table showing 20 trip records with passenger counts: 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14. Working individually (3 minutes), each learner writes a prediction on a sticky note or exercise book: 'On Trip 21, how many passengers will board Kamau's matatu — and can you write a rule that always works?' Learners then share predictions with a neighbour (Think-Pair) for 2 minutes before selected pairs share with the class.	1. Display the Kamau context image (Machakos Country Bus Station, a 14-seater matatu, a conductor with a notebook) and read aloud: 'Kamau runs the Nairobi–Kitengela route every morning. He has kept this tally notebook for 20 trips. Look at his data carefully.' [Point to the table.] 2. Pose the Predict question verbally and in writing on the board: 'On Trip 21, how many passengers will board? Write a prediction AND a rule that will always work.' 3. Say: 'Work silently and individually. You have 3 minutes.' [Start visible timer. Circulate — do NOT hint. Observe what rules learners attempt: e.g., 'always 14', 'alternating', 'average'.] 4. After 3 minutes, say: 'Now turn to your neighbour — share your prediction and your rule. You have 2 minutes.' 5. After pair talk, cold-call EXACTLY 4 pairs using a name-stick draw: 'Pair A — what was your prediction and rule?' [Allow each pair 30 seconds. Write their rules verbatim on the board under the heading: KAMAU'S RULE CANDIDATES.] 6. Ask the whole class: 'Hands up if your rule works perfectly for ALL 20 trips shown.' [Expect no hands, or 1–2 tentative hands.] [WAIT 10 seconds.] 7. Say: 'This is the puzzle we are going to live with for this entire unit. Today is Day 1 of figuring out how Kamau — and YOU — can make smart decisions even when you cannot predict exactly.'	Strategy: PREDICT-BEFORE-INSTRUCTION (Eliciting Prior Conceptions). By asking learners to generate a predictive rule BEFORE any instruction, the teacher surfaces the dominant misconception — that patterns in nature are perfectly deterministic — which is the precise cognitive obstacle that the concept of probability is designed to resolve. Writing rules on the board gives learners concrete claims to test and revise throughout the lesson and unit.	Observable indicator: Do learners attempt a deterministic rule (e.g., 'always 14', 'every 3rd trip is full') or do any learners spontaneously use language like 'sometimes', 'about half the time', 'roughly'? Teacher notes which learners already use probabilistic language — these learners can later be strategically cold-called during the Explain phase. Red flag: if ALL learners say 'impossible to predict anything' — they may be conflating unpredictability with zero mathematical utility, which needs addressing immediately.

Observe Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Analysing Kamau's Notebook (10 min): In pairs, learners receive the full 20-trip tally table and complete a structured analysis worksheet: (i) Count how many trips had a FULL bus (14 passengers). (ii) Count how many trips had FEWER than 14 passengers. (iii) Calculate: 'Out of 20 trips, what FRACTION ended with a full bus?' (iv) Write this fraction as a decimal. (v) Circle any trip-to-trip pattern they notice. Pairs record answers in exercise books. PART B — Coin Toss Experiment (18 min): Each pair receives one 10-shilling coin. Learners set up a two-column tally table in their exercise books: HEADS | TAILS. One learner tosses; the partner records. They perform exactly 30 tosses, recording each outcome with a tally mark. After 30 tosses, each pair calculates: Relative Frequency of Heads = (Number of Heads) ÷ 30. Relative Frequency of Tails = (Number of Tails) ÷ 30. Pairs write their two relative frequencies on a class data strip (a large grid on the board or a shared chart) — one row per pair.	PART A: 1. Distribute the Kamau worksheet. Say: 'In your pairs, answer the five questions on the worksheet. You have 8 minutes. Show all working — no calculators needed for counting.' 2. Circulate actively. Look for: Are learners counting correctly? (Full trips = 8 out of 20, giving 8/20 = 0.4.) Are any pairs noticing the trips with 14 passengers seem to cluster? [Do not correct yet — let them wrestle.] 3. After 8 minutes, cold-call 2 pairs to share their fraction for full-bus trips. Write both fractions on the board. Say: 'Are these the same? Why might two pairs get slightly different answers?' [WAIT 10 seconds.] [Expected: same data, should be same answer — reinforces careful counting.] PART B: 1. Hold up a 10-shilling coin. Ask: 'Before we toss — what are the two possible outcomes?' [Heads / Tails.] 'Does one side have a better chance? How sure are you?' [WAIT 10 seconds. Take 3 responses.] 2. Demonstrate the toss-and-record procedure clearly: 'Toss, let it land on the desk — no catching in the air. Record immediately with a tally mark. Your partner watches and checks. Swap roles every 10 tosses.' 3. Say: 'You have 15 minutes. Begin.' [Circulate — verify pairs are recording each toss immediately, not in batches. Prompt any pair that loses count to restart their current set of 10.] 4. With 3 minutes remaining, say:	Strategy: DATA COLLECTION + ANCHORED COMPARISON. The two-part observe phase is intentionally sequenced: learners first study REAL-WORLD messy data (Kamau's 20 trips) before they generate their own controlled experimental data (coin tosses). This parallel structure — both datasets showing variation around an expected value — is the core sensory anchor for the concept of experimental probability. The class grid on the board makes the convergence effect visible and social, not just numerical.	Observable indicators: (1) Can each pair correctly compute the fraction of full-bus trips from Kamau's data? (Expected: 8/20 = 0.4. Accept equivalent fractions.) Flag any pair writing a percentage without a fraction — they may lack fraction fluency. (2) After coin tosses, are all pairs' relative frequencies between 0.33 and 0.67? Values outside this range may indicate miscounting — ask the pair to recount their tallies. (3) On the class grid, scan for any pair claiming exactly 0.50 — ask them to double-check, as this is statistically unlikely with only 30 tosses and may indicate fabricated data.

		'Stop tossing. Calculate your two relative frequencies now — number of heads divided by 30, number of tails divided by 30. Write them as decimals to 2 decimal places.' 5. Direct each pair to write their Heads relative frequency on the class grid on the board. [You will now have ~15–20 data points from ~10 pairs.]		
Explain Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners examine the completed class grid of relative frequencies from all pairs. The teacher guides learners to calculate the CLASS TOTAL (all heads across all pairs ÷ total tosses by all pairs). Learners observe that the class total is closer to 0.5 than most individual pair results. In pairs, learners then write a two-sentence explanation connecting this finding back to Kamau's notebook: 'How does what we found with the coin help explain why Kamau's data shows 8 full trips out of 20 — not exactly 10?' Pairs share their explanations. The teacher introduces the formal vocabulary: EXPERIMENTAL PROBABILITY and RELATIVE FREQUENCY, anchored directly to both datasets.	1. Point to the class grid. Ask: 'Look at everyone's relative frequencies. What do you notice about them — are they all the same?' [WAIT 10 seconds. Cold-call 3 learners.] Expected responses: 'They are all different', 'Some are above 0.5, some below', 'None is exactly 0.5.' 2. Say: 'Let us now calculate what happens when we combine EVERYONE's results.' Lead the class to sum all heads and all tosses: write on the board 'CLASS TOTAL = ___ heads out of ___ tosses = ___.' Ask one learner to come up and compute the decimal. [With ~10 pairs × 30 tosses = 300 tosses, expected ≈ 147–153 heads, giving ≈ 0.49–0.51.] 3. Ask: 'Why is the class total closer to 0.5 than most individual pairs?' [WAIT 15 seconds. This is a KEY thinking moment — resist filling the silence.] Cold-call 4 learners. Write key phrases from their answers on the board. 4. Bridge back to Kamau: Say: 'Now — here is the link to Kamau. If the theoretical probability of a full bus is, say, 0.4 — based on his data — should we expect exactly 8 full trips in every 20? Or sometimes 7, sometimes 9,	Strategy: CONVERGENCE DEMONSTRATION + DISCIPLINARY VOCABULARY ANCHORING. The teacher strategically uses the pooled class data to make the Law of Large Numbers tangible BEFORE naming it. Learners see the convergence happen on the board with their own numbers. Vocabulary (experimental probability, relative frequency) is introduced AFTER learners have already experienced the concept, so the term labels a lived experience rather than arriving as an abstraction. The explicit bridge back to Kamau's notebook prevents the coin activity from becoming a decontextualised exercise.	Observable indicators: (1) Can learners correctly apply the formula P(event) = frequency ÷ n to both datasets without prompting? (2) In the two-sentence written explanation, do learners use causal language — 'because each trip is separate', 'because chance varies', 'because more trials get closer'? Collect 4–5 exercise books at random to read these explanations — they reveal depth of conceptual understanding. (3) Red flag: any learner who writes 'Kamau should get exactly 8 full buses every 20 trips' has not yet grasped variability — note this learner for targeted follow-up in Lesson 2.

		sometimes 10?' [WAIT 10 seconds. Take 3 responses.] 5. Say: 'What Kamau found with his matatu, and what we found with our 10-shilling coin, has a mathematical name.' Write on the board: EXPERIMENTAL PROBABILITY = Number of times event occurs ÷ Total number of trials Also written as: P(event) ≈ frequency / n 6. Ask: 'In Kamau's notebook, what is the experimental probability of a full bus?' [8 ÷ 20 = 0.4.] Cold-call one learner to compute it live. 7. Ask: 'What is the experimental probability of heads from our CLASS data?' [Class total ÷ total tosses.] Cold-call a different learner. 8. Say: 'The mathematical word for this fraction — outcomes we observed ÷ total trials — is RELATIVE FREQUENCY. Experimental probability and relative frequency are the same thing. Write that definition in your exercise books now.' 9. Pose the synthesis question: 'In pairs, write two sentences: How does the coin experiment help explain why Kamau gets 8 full buses — not exactly 10 — in 20 trips?' Give 4 minutes. Cold-call 3 pairs to read their sentences aloud.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Regrouping whole number place values Source: Khan	Learners receive two sticky notes (or small paper slips) each. On the FIRST slip, they write one question that today's lesson has raised for them — something they still do not understand or want to explore. On the SECOND slip, they write one thing they now understand about	1. Introduce the DQB: 'This board will travel with us for the entire probability unit. Every lesson, we will add to it and answer questions from it. Right now, Kamau has given us a puzzle — and we have started solving it. But I am sure today raised MORE questions than	Strategy: DRIVING QUESTION BOARD — QUESTION GENERATION. The DQB serves as a public record of the class's collective intellectual curiosity. By reading questions aloud and mapping them to future lessons, the teacher demonstrates that	Observable indicators: (1) Quality of QUESTION slips — are learners asking surface questions ('What is probability?') or mechanistic/causal questions ('Why does more data give a better estimate?')? The latter indicates readiness for Lesson 2's deeper exploration of

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Kamau's problem that they did not understand at the start of the lesson. Both slips are posted on the class DQB, which is divided into two columns: QUESTIONS WE STILL HAVE | THINGS WE NOW UNDERSTAND. The teacher reads out selected questions and categorises them aloud, foreshadowing upcoming lessons.	it answered.' 2. Distribute two slips per learner. Say: 'Slip 1 — write ONE question you still have. It can be about Kamau, about the coin, about probability, about anything from today. Slip 2 — write ONE thing you understand NOW that you did not understand at the START of this lesson. You have 3 minutes. Write in full sentences.' 3. Collect slips by row. Read 6–8 QUESTION slips aloud (select a range: simple, deep, surprising). As you read each one, say: 'We will answer this in Lesson ___' and write the lesson number next to it on the board. Likely questions to anticipate and address: — 'Why doesn't experimental probability exactly equal what we expect?' → Lesson 1 (partial answer today; deeper in Lesson 2) — 'What if we tossed the coin 1000 times — would it be exactly 0.5?' → Lesson 2 — 'How do you work out the probability of something when you cannot do an experiment?' → Lesson 3 (theoretical probability, sample space) — 'Can two things happen at the same time — like a full bus AND a half-empty bus?' → Lesson 5 (mutually exclusive events) 4. Read 3–4 UNDERSTANDING slips aloud. Say: 'Listen — your classmates are already making sense of Kamau's puzzle.' 5. Post all slips on the DQB. Say: 'This board is our intellectual home for the next 12 lessons. Every time we answer a question from this board, we will flip that slip over or mark it SOLVED.'	learner questions genuinely shape the unit's direction and that confusion is productive, not shameful. The two-slip structure (question + understanding) prevents the DQB from becoming purely a record of ignorance — it also celebrates growing knowledge from the very first lesson.	experimental vs theoretical probability. (2) Quality of UNDERSTANDING slips — do learners correctly reference the phenomenon ('I understand that Kamau cannot predict one trip but can estimate over many trips') or do they restate a procedure ('I understand how to calculate relative frequency')? Conceptual understandings are the target. Teacher photographs all DQB slips for lesson planning reference.

Model Building Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners draw their FIRST version of a class model in their exercise books. The model is a simple annotated number line from 0 to 1, on which they place and label two events from today's lesson: (i) Kamau's experimental probability of a FULL BUS (8/20 = 0.4) and (ii) their OWN experimental probability of HEADS from the coin toss (their pair's value). Learners annotate the number line with three labels: IMPOSSIBLE (0), CERTAIN (1), and EVEN CHANCE (0.5). They then write one sentence predicting: 'By the end of this unit, I think the model will change/grow by ___.' Models are left in exercise books — learners will revise them in every subsequent lesson.	1. Say: 'Every scientist and mathematician builds a model — a picture or diagram that captures what they understand so far. Today, we will build the first version of our probability model. It will grow every lesson.' 2. Instruct: 'Draw a horizontal line across a full page. At the left end, write 0. At the right end, write 1. This is our PROBABILITY SCALE.' 3. Say: 'Place three anchor labels: write IMPOSSIBLE directly below 0, CERTAIN directly below 1, and EVEN CHANCE directly below 0.5. Mark these with small vertical tick marks.' 4. Say: 'Now place TWO events on your number line. First: Kamau's experimental probability of a full bus — where does 0.4 sit? Mark it with a dot and label it: P(Full Bus, Kamau's data).' 5. Say: 'Second: place YOUR pair's experimental probability of heads. Mark it with a different coloured dot or a triangle, and label it: P(Heads, our experiment).' 6. Say: 'Look at where your two dots sit. Are they on the same side of 0.5? What does that tell you?' [WAIT 10 seconds. Take 2 responses.] 7. Say: 'Finally — in one sentence below your number line, predict: By the end of this unit, how do you think this model will grow or change? What new things might we add to it?' Give 2 minutes for writing. 8. Cold-call 3 learners to read their prediction sentences. Affirm all reasonable predictions. Say: 'We will come back to this page in every lesson. Do NOT erase anything — add to it, annotate it,	Strategy: CUMULATIVE MODEL REVISION (Probability Number Line). The number line model is a low-threshold, high-ceiling tool: every learner can immediately place two values on it, but the model has headroom to grow as learners add theoretical probabilities, mutually exclusive event pairs, combined event probabilities, and tree diagram outcomes in later lessons. By instructing learners never to erase — only annotate — the teacher makes the revision process visible and validates the idea that scientific models are always incomplete and growing.	Observable indicators: (1) Is each learner's 0.4 dot placed correctly — to the LEFT of 0.5? A dot placed at or above 0.5 indicates a calculation error or conceptual confusion about the fraction 8/20. (2) Is the learner's personal heads probability placed as a dot distinct from the Kamau dot? Learners who place only one dot may not have computed their own relative frequency. (3) Prediction sentences: are learners anticipating new types of events ('we might add probability of two things happening'), new methods ('maybe we will learn a formula'), or do they predict only numerical refinement ('the numbers will be more accurate')? The former indicates readiness for the sample space and combined events lessons ahead. Collect 5 exercise books at lesson close for review.

		but never erase. Your model is a record of your growing thinking.'		

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions in your professional journal: 1. PHENOMENON LAUNCH: Did learners genuinely find Kamau's tally notebook puzzling, or did some dismiss the phenomenon as 'obvious'? If the latter, how will you increase the cognitive hook — e.g., by adding a bet ('I will give anyone 10 shillings who can write a rule that predicts all 20 trips correctly') to raise the stakes of the prediction phase? 2. COIN TOSS QUALITY: Were any pairs getting suspiciously round numbers (exactly 15 heads, exactly 15 tails)? If so, were they fabricating data to 'look correct'? This is a values and integrity issue — address it in the next lesson by reinforcing that real experimental data is SUPPOSED to be imperfect, and that honesty in recording is a mathematical virtue. 3. DQB RICHNESS: Count the proportion of CAUSAL questions ('Why...?', 'How...?') versus DEFINITIONAL questions ('What is...?') on the DQB. If more than 60% are definitional, learners may not yet be thinking like probability investigators. Plan a brief 5-minute 'question upgrade' activity at the start of Lesson 2 to model turning definitional questions into causal ones. 4. CONVERGENCE MOMENT: Did learners notice the convergence of the class pooled total toward 0.5? If this was not striking to them — because the class total happened to land at 0.48 or 0.52 — consider how to dramatise the effect in Lesson 2 by adding a simulation of 1000 coin tosses using mobile phones or a chart-based simulation, so the convergence is visually overwhelming. 5. MODEL DIVERSITY: Review the collected exercise books. Are learners' number line placements accurate? Errors here will compound in Lesson 3 when theoretical probability is introduced. Prepare a brief peer-checking activity for the start of Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Kamau's 20-trip matatu tally notebook showing passenger counts (6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14) with no single rule that perfectly predicts every trip; our own coin-toss experiment showing that 30 tosses of a 10-shilling coin produced different numbers of heads and tails for each pair, with no pair getting exactly 15 of each, yet the class pooled total was close to 0.5.
What did I learn?	Experimental probability (relative frequency) = number of times an event occurs ÷ total number of trials. Individual experiments vary — Kamau's 20 trips gave P(Full Bus) = 8/20 = 0.4, and each pair's coin experiment gave a slightly different P(Heads). However, when we pooled all pairs' results, the class total moved closer to 0.5. This shows that MORE trials give a more reliable estimate of probability, even though no single trial can be predicted with certainty.
How does this explain the phenomenon?	Kamau cannot predict whether a single seat will be filled because each passenger's decision to board is an independent, uncertain event — just like each coin toss is independent. However, over many trips (or many tosses), the FRACTION of times an event happens settles toward a stable value. This stable fraction IS the experimental probability, and it gives Kamau — and us — a mathematical tool to plan and make decisions despite uncertainty, even when we cannot predict any single outcome.

LESSON 2 (40 minutes): The Probability Scale — Placing Kamau's World on a Number Line from 0 to 1	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners formalise the probability scale $0 \le P(E) \le 1$ by classifying real Kenyan events from impossible to certain and anchoring every point back to Kamau's tally notebook.
Knowledge	– Define impossible events ($P(E) = 0$), certain events ($P(E) = 1$), and events in between ($0 < P(E) < 1$). – State the rule $0 \le P(E) \le 1$ for any event E. – Explain why Kamau's relative frequencies from Lesson 1 must always fall within this range.
Skills	– Place a given event correctly on a 0-to-1 probability number line with written justification. – Convert Kamau's tally frequencies into probability values and verify they satisfy $0 \le P(E) \le 1$.
Attitudes	– Appreciate that mathematical language gives Kamau a precise and fair tool for describing uncertainty. – Show responsibility by justifying classifications with evidence rather than guessing.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	Having established in Lesson 1 that experimental probability approaches a stable value over many trials, learners now formalise the boundaries of all probability values — anchoring every frequency from Kamau's notebook to a universal scale and setting up the concept of sample space in Lesson 3.
Safety Notes	No specific hazards. Ensure printed event-cards are handled carefully and not torn during the sorting activity.

B. LESSON OVERVIEW

In Lesson 1, learners discovered that Kamau's 20-trip tally gave an experimental probability of P(Full Bus) = 8/20 = 0.4, and that pooling more trials moves the estimate toward a stable value. But what does 0.4 actually mean on a bigger mathematical map? Lesson 2 zooms out to that map — the probability scale — by asking learners to sort 12 vivid Kenyan real-life events from obviously impossible to obviously certain, then locate every event on a shared 0-to-1 number line drawn on the board. Events range from the absurd ("a 14-seater matatu carries 20 passengers") through the highly likely ("it will rain in Kisumu in April") to the guaranteed ("the matatu will eventually stop at a stage"). By physically placing event-cards on the number line and defending each placement, learners build an embodied, social understanding of the scale before the formal inequality $0 \le P(E) \le 1$ is introduced.\n\nThe Explain phase then bridges back to Kamau's notebook: every frequency ratio from Lesson 1 is verified to sit inside the scale, and learners compute two new probability estimates from the tally data — P(bus carries fewer than 8 passengers) and P(bus is completely full) — confirming both land between 0 and 1. This grounds the abstract rule in familiar evidence and reinforces that Kamau's notebook is a mathematical object, not just a record.\n\nThe lesson closes with an updated Driving Question Board where learners add the new question "Can we ever be 100% sure about Kamau's next trip?" and revise their cumulative probability model to show the scale as a labelled number line with Kamau's key probabilities marked. By the end, learners can articulate why no entry in Kamau's notebook can ever give a probability less than 0 or greater than 1 — and why that boundary itself is mathematically powerful.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Video: Make a Ten] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a sticky note and is asked to write down one event from Kamau's matatu world (or from their own daily life in Nairobi, Kisumu, or Kericho) that they believe is IMPOSSIBLE and one event they believe is CERTAIN. They also estimate where P(Full Bus) = 0.4 from Lesson 1 would sit if impossible = 0 and certain = 1. Learners record both events and the estimated position in their exercise books before any discussion.	Display Kamau's tally notebook data on the board (the 20-trip list from the phenomenon). Ask: 'Looking at Kamau's notebook from last lesson, think quietly — what is one thing about his matatu that could NEVER happen, and one thing that is GUARANTEED to happen? Write both in your book right now. You have 90 seconds — go.' [Wait 90 seconds — circulate silently, read 3–4 responses over shoulders.] After writing, ask: 'Without telling your neighbour, also write a number between 0 and 1 for where you think P(Full Bus) = 8/20 sits on a line from impossible to certain.' [Wait 20 seconds.] Cold-call 3 learners: 'Achieng, read me your impossible event.' / 'Mwangi, your certain event.' / 'Fatuma, your number for 8/20.' Do NOT correct — record all three on the board under the heading 'Our Starting Ideas'.	Predict-Before-Instruction (PBI): By committing predictions to paper before any teaching, learners activate and expose prior conceptions about certainty and impossibility. Recording Kamau-specific examples immediately connects the abstract task to the anchoring phenomenon and creates a written baseline the teacher can compare against end-of-lesson understanding.	Teacher reads 3–4 sticky notes/books during the 90-second writing period. Look for: (1) whether learners confuse 'unlikely' with 'impossible'; (2) whether the number chosen for 8/20 is outside 0–1 (revealing a misconception to address in Explain). Flag any learner who writes a negative number or a number > 1 — these learners need targeted questioning in the Explain phase.
Observe Phase 📹 **VIDEO:** [Video: Make a Ten] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In groups of 3, learners receive a set of 12 printed event-cards (see Teacher Template below) showing Kenyan real-life scenarios. Groups place the cards in order along a rope or chalk line on the floor labelled 0 at one end and 1 at the other, physically walking cards to their chosen positions. After placing all 12 cards, each group sketches the resulting number line in their exercise books, marking each event with a letter (A–L). Groups then share one placement they found difficult to agree on. The 12 events are: A — A 14-seater matatu in Nairobi carries exactly 20 seated passengers on one trip. B — A mandazi bought at a Gikomba market stall costs exactly 0 shillings. C — The sun rises tomorrow morning in Nairobi.	Distribute the event-card sets and the rope/chalk line. Say: 'Your group's job is to be probability detectives. Place each card exactly where you think it belongs on the line from 0 — impossible — to 1 — certain. You must agree as a group AND be ready to explain WHY. You have 8 minutes.' [Set a visible timer. Circulate and listen — do NOT confirm or deny placements.] At 4 minutes, prompt groups that are stuck: 'For cards G and K, Kamau gave us data. Can you use his 20 trips to calculate a number?' [Wait 15 seconds before moving on.] After 8 minutes, call time. Ask one spokesperson per group to share the one card their group disagreed most about. Record disagreed-upon cards on the board. Then ask the whole	Embodied Sorting (Card-Line Activity): Physical placement of event-cards on a spatial number line leverages kinesthetic and visual learning to make the abstract probability scale concrete. Disagreement within groups is intentional — productive conflict around cards E (Kisumu rain — high but not certain) and G/K (data-dependent) forces learners to distinguish 'feels likely' from 'is mathematically definable'. Sketching the final line in exercise books consolidates the experience as a personal record.	During circulation, note: (1) Do groups place cards A, B, J at exactly 0 and cards C, D, I at exactly 1 without prompting? (2) Do any groups place card F (fair coin) significantly away from 0.5? (3) Do groups use Kamau's data (8 full trips out of 20 = 0.4 for G; 4 trips with fewer than 8 passengers out of 20 = 0.2 for K) or rely on gut feeling? Groups that do not use the data for G and K need the bridging prompt during circulations. Record group placements on a quick sketch — use these to structure the Explain phase discussion.

	D — Kamau's matatu carries at least 1 passenger on a busy Monday morning. E — It rains in Kisumu at some point during the month of April. F — A fair coin flipped once lands on heads. G — Kamau's matatu is completely full (14 passengers) on a randomly chosen trip (use his notebook). H — A farmer in the Rift Valley harvests maize in a season with zero rainfall. I — A learner chosen at random from a class of 30 is either a boy or a girl. J — A die rolled once shows a 7. K — Kamau's matatu carries fewer than 8 passengers on a randomly chosen trip (use his notebook). L — A githeri pot left on a charcoal jiko overnight with no water eventually burns.	class: 'Which two cards are you MOST confident belong at exactly 0? Raise hands.' Cold-call 2 learners for justification: 'Otieno, why exactly 0 for card A?' / 'Njeri, is there any world in which card J could happen — roll a standard die and get 7?' [Wait 10 seconds each.] Do not yet give answers — hold tension for the Explain phase.		
Explain Phase 📹 **VIDEO:** Video: Make a Ten Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher draws a large 0-to-1 number line on the board and invites groups to place the agreed class positions for all 12 cards. Learners then copy the formal rule $0 \leq P(E) \leq 1$ into their books, label the three landmark zones (impossible = 0; certain = 1; uncertain = between 0 and 1), and work through three guided calculations using Kamau's tally: (1) P(Full Bus) = 8/20 = 0.4 (verified from Lesson 1); (2) P(Fewer than 8 passengers) = 4/20 = 0.2; (3) P(Not a full bus) = 12/20 = 0.6. Learners mark all three values on their personal number line and write one sentence explaining why none of them can be less than 0 or greater than 1.	Build the class number line on the board by asking each group in turn: 'Group 2 — where did you place card C? Come and mark it.' Work through all 12 cards with brief justifications. When the class reaches card A (matatu carries 20 when it seats 14), say: 'This is physically impossible. A 14-seater matatu cannot legally or physically seat 20 passengers. So P(A) = exactly 0. What do we call an event like this?' [Wait 10 seconds.] Cold-call: 'Kipchoge, give me the word.' [Answer: impossible.] Write IMPOSSIBLE = P(E) = 0 on the board. Repeat the pattern for card C: 'The sun rising in Nairobi tomorrow — can we give this a probability of 1.5? Why not?' [Wait 10 seconds.] Cold-call: 'Wanjiku, why can no probability be more than 1?' Then formally write: $0 \leq P(E) \leq 1$ on the board inside a box. Say: 'This is the law of the probability scale. Every single	Anchor-Back Calculation (Phenomenon Link): Every abstract concept — the inequality, the labels impossible/certain/uncertain — is verified immediately using numbers from Kamau's tally. This 'anchor-back' move prevents the probability scale from feeling like a disconnected rule and instead positions it as a description of data learners already own from Lesson 1. The thumbs-up/down check after each calculation provides rapid whole-class formative feedback with low stakes.	Look for: (1) Learners who give a thumbs-down when 0.4 satisfies the inequality — these learners may be misreading the direction of the inequality and need individual follow-up. (2) During the 60-second pair discussion, circulate and listen for learners who suggest that if Kamau had 0 full trips the probability would be 'negative' — a key misconception to address immediately: 'Zero full trips means P = 0/20 = 0, which is still inside the scale, not below it.' (3) Check that the written sentence in exercise books includes both boundary conditions (not just 'less than 1' but also 'at least 0').

		number in Kamau's tally — every frequency ratio we calculated last lesson — must obey this rule. Let us check.' Model the three Kamau calculations on the board step by step. After each, ask: 'Does 0.4 obey $0 \leq P(E) \leq 1$? Yes or no — thumbs up or down.' After all three, pose the challenge: 'Can you find any trip-result in Kamau's notebook that would give us a probability outside 0 to 1? Discuss with your partner for 60 seconds.' [Wait 60 seconds.] Cold-call 2 pairs: 'Amina, what did you find?' / 'Baraka, do you agree?' Confirm: 'No — because the number of times something happens can never be less than 0 or more than the total number of trials. That is why the rule is universal.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Make a Ten] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board posted on the wall. They review the questions added in Lesson 1 (e.g., 'Why do Kamau's tallies never give exactly the same number twice?'). Each learner writes one new question or updated understanding on a sticky note prompted by today's lesson. The class then collectively decides where to post a new anchor question: 'Can we ever be 100% sure about Kamau's next trip — and what would P(E) = 1 actually mean for his route?' Learners also mark which Lesson 1 questions have now been partially answered by today's scale.	Point to the DQB and say: 'Last lesson we asked why Kamau's counts keep changing. Today we gave every possible answer a home on a scale from 0 to 1. That raises a new question I want YOU to write.' Distribute sticky notes. Say: 'Write ONE question that today's lesson made you think about — something you are still not sure of about Kamau's probability. It can be about the scale, about the number 1, about whether any event in Kenya is truly certain. 60 seconds — write now.' [Wait 60 seconds.] Ask 4 learners to read their questions aloud — cold-call: 'Chebet, read yours.' / 'Onyango, yours.' / 'Muthoni, yours.' / 'Hassan, yours.' After each, ask the class: 'Thumbs up if your question is similar.' Cluster similar questions together on the DQB. Then point to the pre-written anchor question 'Can we ever be 100% sure about Kamau's next	Driving Question Board — Question Clustering: The DQB is a living record of the class's collective sensemaking journey. By physically clustering similar student questions, the teacher makes visible the shared intellectual puzzles of the room and validates every contribution. Marking Lesson 1 questions as 'partially answered' (not 'done') models scientific thinking — understanding builds incrementally, not in one lesson.	Read all posted sticky notes after class. Categorise questions into three types: (1) Boundary questions ('What if P = 0.5 exactly — is that special?') — shows readiness for Lesson 4 on mutually exclusive events. (2) Data questions ('What if Kamau had 100 trips of data?') — shows readiness for deeper work on experimental vs theoretical probability. (3) Misconception questions ('Can P(E) be 1.0 for Kamau if every seat is always full?') — these learners need targeted correction at the start of Lesson 3. Use this categorisation to inform the opening of Lesson 3.

		trip?' and say: 'I am adding this question to our board today. We will answer it more fully as we discover sample space and the laws of probability in the coming lessons.' Mark Lesson 1 questions about experimental variability with a green star to show they are 'partially answered'.		
Model Building Phase 📹 **VIDEO:** Video: Make a Ten Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open to their cumulative probability model in their exercise books (started in Lesson 1 as a sketch of Kamau's notebook with a frequency bar). They now ADD a clearly labelled 0-to-1 number line below the notebook sketch. On this line, they mark and label at least four points: P(impossible event) = 0; P(Full Bus from Kamau's data) = 0.4; P(Not Full Bus) = 0.6; P(certain event) = 1. They also write one 'model update sentence': 'Every probability value from Kamau's notebook is a fraction between 0 and 1, showing how likely each outcome is — 0 means it cannot happen, 1 means it must happen, and Kamau's bus being full (0.4) sits closer to unlikely than to certain.' Learners who finish early add a fifth point: P(Fewer than 8 passengers) = 0.2.	Say: 'Open to your Kamau model from Lesson 1. We are going to upgrade it right now. Under your notebook sketch, draw a horizontal number line from 0 to 1 — leave about 6 cm of space. Mark 0 on the left, 1 on the right, and 0.5 in the middle.' [Pause 30 seconds — draw the same line on the board.] 'Now mark these four points using the data we worked with today' — read the four points slowly while pointing to each on the board version. 'Finally, write the model update sentence I am putting on the board now.' [Write the sentence on the board.] 'You have 3 minutes. This is silent individual work — it is your personal model.' [Circulate during 3 minutes — look at 6–8 books, note accuracy of fraction placements.] At 3 minutes, call: 'Swap books with your neighbour. Check: is their 0.4 closer to 0 than to 1? Is their 0.6 correctly on the right side of 0.5? Give one specific written comment.' [Allow 90 seconds for peer check.] Close: 'Your model now has two layers — Kamau's raw data AND the scale that governs all probability. Next lesson we will add a third layer: every possible outcome listed in a sample space.'	Cumulative Model Revision (Layered Modelling): Building the model across lessons in a single exercise-book location creates a tangible, visual record of growing understanding. Each lesson adds a new 'layer' (Lesson 1: experimental frequency; Lesson 2: probability scale; Lesson 3: sample space). The peer-check move embeds immediate low-stakes assessment and reinforces precision in placing fractions on a number line — a key mathematical skill that underpins all subsequent probability work.	During circulation, check: (1) Is 0.4 placed to the left of 0.5 (closer to 0)? Misplacement suggests the learner does not yet have a secure sense of fraction magnitude. (2) Is 0.6 placed to the right of 0.5? (3) Does the model update sentence include both 0 and 1 as boundary cases? Collect 5–6 books at the end of class for a quick marking scan — use a green tick for correct placement of all four points and a yellow dot for any misplaced fraction, to return at the start of Lesson 3.

D. TEACHER REFLECTION

1. During the card-sorting activity, did all groups use Kamau's tally data (8/20 and 4/20) to place cards G and K, or did most groups rely on intuition? If intuition dominated, how will I build a stronger bridge between raw frequency data and the probability scale in Lesson 3?\n2. Which of the 12 event-cards generated the most disagreement, and does that disagreement reveal a specific misconception (e.g., confusing 'very unlikely' with 'impossible') that I need to address explicitly at the start of Lesson 3?\n3. When I introduced the formal inequality $0 \leq P(E) \leq 1$, did learners connect both boundary conditions ($P = 0$ AND $P = 1$) with equal confidence, or did they find one boundary more intuitive than the other? What adjustment to the Explain phase would help the less-intuitive boundary?\n4. Looking at the DQB sticky notes collected today, what is the most common type of student question — boundary questions, data questions, or misconception questions — and how does this inform the opening hook I will use in Lesson 3?\n5. In the Model Building phase, how many learners misplaced 0.4 or 0.6 on their number lines during the peer-check? If more than one-quarter of the class showed fraction-placement errors, do I need to insert a brief number-line fluency warm-up at the start of Lesson 3 before moving to sample space?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We sorted 12 Kenyan real-life events — from a matatu carrying 20 passengers in a 14-seater (impossible) to the sun rising tomorrow in Nairobi (certain) — onto a number line from 0 to 1. We also placed Kamau's calculated probabilities (P(Full Bus) = 0.4, P(Not Full) = 0.6, P(Fewer than 8) = 0.2) on the same line.
What did I learn?	Every probability must satisfy $0 \leq P(E) \leq 1$. An impossible event has $P(E) = 0$ and a certain event has $P(E) = 1$. All of Kamau's frequency ratios from his 20-trip notebook fall inside this range — none can be negative or greater than 1 — because the count of an outcome can never be less than 0 or more than the total number of trials.
How does this explain the phenomenon?	Kamau cannot predict with certainty whether the next seat will be filled because the probability of a full bus is 0.4 — it sits between 0 and 1, meaning it is neither impossible nor guaranteed. The probability scale shows mathematically that Kamau's uncertainty is real and bounded: he will never face an outcome with $P < 0$ or $P > 1$, so even though he cannot be 100% sure, he can say precisely how sure he is.

LESSON 3 (80 minutes): Probability Space (Sample Space): Listing Every Possibility for Kamau's Route	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' ability to systematically generate and use a probability space (sample space) as the mathematical foundation for assigning theoretical probability to real-world uncertain events.
Knowledge	Learners will know that: (a) a sample space is the complete set of all possible outcomes of a random experiment; (b) theoretical probability of an event = (number of favourable outcomes) ÷ (total number of outcomes in the sample space); (c) a two-way table (grid) is an efficient tool for listing the sample space of two-stage experiments such as rolling two dice or recording seat status at two stops.
Skills	Learners will be able to: (a) generate the probability space of different events systematically using lists and two-way tables; (b) determine theoretical probability of events from a constructed sample space; (c) connect the size and structure of the sample space to why Kamau cannot perfectly predict outcomes but can quantify his chances mathematically.
Attitudes	Learners will appreciate that systematic listing — leaving nothing out and counting nothing twice — is an act of mathematical integrity and is directly useful for making fairer, safer, and smarter decisions in everyday Kenyan life.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	In Lesson 1 learners met Kamau's puzzle and explored experimental probability from his tally notebook. In Lesson 2 they established the range and meaning of probability values. Now in Lesson 3 learners confront the question: before Kamau even runs a trip, how many distinct outcomes are possible? By building the sample space for two stops (Mlolongo and Athi River) and extending to a 6×6 dice grid at a school game stall, learners discover that theoretical probability flows directly from a complete, unambiguous list of possibilities — answering the DQB question: 'How do we count all possibilities without missing any?'
Safety Notes	No physical hazards. If physical dice are used, remind learners to roll on the desk surface only and collect dice promptly to avoid distraction or loss. Ensure learners with visual impairments receive enlarged printed copies of the two-way table grid.

B. LESSON OVERVIEW

Learners open by revisiting Kamau's tally data from previous lessons and are challenged to think about ALL possible outcomes — not just what happened, but what could happen. In the Predict phase they attempt to list every possible boarding outcome for two consecutive stops without guidance. In the Observe phase they work in pairs to construct the sample space {TT, TE, ET, EE} for Kamau's two-stop model using a structured two-way table, then extend this to rolling two dice using a 6×6 grid (connecting to a school game stall at Athi River Secondary School). In the Explain phase they derive theoretical probability from the sample space and compare it to the experimental estimates from Lesson 1. The DQB is updated with the new anchoring question and a revised class model is produced showing a sample space diagram linking back to Kamau's route.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each learner receives Kamau's simplified two-stop scenario card	Distribute scenario cards (or direct learners to the ARES screen).	Individual written prediction followed by partner comparison.	Teacher scans books during individual think time. Look for: (1)

📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Sample Spaces and Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(printed or on the ARES device): 'At Mlolongo stop, a seat is either Taken (T) or Empty (E). At Athi River stop, the same seat is either Taken (T) or Empty (E). In your exercise book, list every possible pair of outcomes for these two stops. How many pairs do you think there are? Write your prediction and your list before we do anything else.' Learners work individually and silently for 4 minutes, writing their lists. They then share with a shoulder partner for 2 minutes to compare — do they have the same outcomes? Any disagreements?	Say: 'Do not help each other yet — I want YOUR thinking first. You have 4 minutes. Go.' Circulate and scan books without commenting. After 4 minutes say: 'Turn to your shoulder partner and compare lists. Do you agree on how many outcomes there are? You have 2 minutes.' After partner talk, cold-call 3 learners (one who listed 2 outcomes, one who listed 3, one who listed 4) to share their lists on the board. Ask each: 'How did you decide you had found them all?' Allow 10 seconds of wait time after each question before accepting a response. Record all three versions on the board under the heading 'Our Predictions' — do NOT correct yet.	This activates prior knowledge about outcomes and surfaces the key misconception — many learners list only 2 or 3 outcomes because they forget that order matters (T then E is different from E then T in a sequence of stops). Keeping wrong predictions visible on the board creates productive cognitive dissonance that the Observe phase will resolve.	learners who write only {TT, EE} — missing mixed outcomes; (2) learners who write {TT, TE, EE} — missing ET; (3) learners who correctly list all four {TT, TE, ET, EE}. Mentally note the distribution — this informs cold-call selection and the depth of scaffolding needed in Phase 2.
Observe Phase 📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Sample Spaces and Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Kamau's Two-Stop Sample Space (25 min): Working in pairs, learners receive (or draw) a blank two-way table with 'Stop 1: Mlolongo' across the top (columns: T, E) and 'Stop 2: Athi River' down the side (rows: T, E). They fill in each cell with the combined outcome (e.g., row T / col T → TT). They list the four outcomes as a set: {TT, TE, ET, EE}. They answer three questions on the ARES activity card: (i) How many outcomes are in the sample space? (ii) In how many outcomes is at least one seat Taken? (iii) What is the theoretical probability that both seats are Empty? They record answers in their books. PART B — School Game Stall Extension (20 min): Learners are told: 'At the Athi River Secondary School end-of-term fundi fair, a game stall uses two standard dice. A player wins a prize if both dice	PART A: Distribute or display the blank two-way table. Say: 'Fill in every cell — no cell should be blank. If you and your partner disagree on what goes in a cell, write both options and we'll discuss.' After 8 minutes, cold-call one pair to read out their completed table. Ask the class: 'Does everyone's table match? If not, raise your hand.' Address discrepancies immediately. Then ask the three questions aloud, giving 10 seconds of wait time each. Cold-call 3 different pairs for answers. Ask: 'How does this table prove that our prediction of [X outcomes] was or wasn't complete?' PART B: Say: 'Now we scale up. Same idea, bigger table. Take 12 minutes to fill your 6×6 grid — every cell.' Circulate; if pairs miss cells, tap the blank cell and say: 'Is this stop possible? What would go	Two-way table construction (structured listing). By physically filling in every cell of a grid, learners embody the multiplication principle without being told it abstractly. The constraint of 'no blank cells' forces exhaustive enumeration. Connecting the 2×2 table to the 6×6 table helps learners generalise the method. The fairness question about the game stall activates critical thinking and connects mathematics to social justice (KICD value: social justice).	Observe whether pairs fill ALL 36 cells in the dice grid or leave cells blank (indicates incomplete enumeration). Check whether learners correctly identify 6 winning outcomes (diagonal of the grid) and compute $6/36 = 1/6$. Listen for whether learners can articulate WHY the two-way table guarantees completeness. Target pairs who struggled in Part A for additional questioning during Part B.

	show the same number. Let's build the full sample space.' Each pair draws a 6×6 grid. The columns are labelled Die 1: 1–6; rows are Die 2: 1–6. They write all 36 ordered pairs in the cells. They then: (i) circle all outcomes where both dice show the same number; (ii) count the favourable outcomes; (iii) calculate the theoretical probability of winning. They compare this to the question: 'Is this game at the fundi fair fair to the player? Why or why not?' PART C — Link to Kamau (5 min): Pairs answer: 'How is building this 6×6 grid the same kind of thinking as building Kamau's 2×2 table? What would Kamau's sample space look like if he had THREE stops — Mlolongo, Athi River, and Kitengela town?'	here?' After 12 minutes, display a completed reference grid on the board (pre-drawn or projected). Ask: 'How many outcomes did you get in total?' Accept answers aloud. Then: 'Circle the winning outcomes — where both dice match.' Give 3 minutes. Cold-call 2 pairs to state their circled outcomes and count. Write the probability fraction on the board: 6/36. Simplify together: 1/6. Ask: 'Is this game fair? Give me a yes-or-no with a reason. I'll wait 10 seconds then I'll pick someone.' Cold-call 2 learners. PART C: Pose the three-stop question. Give 3 minutes of pair discussion. Cold-call 1 pair. Do NOT solve it — say: 'Hold that thought for our model-building phase.'		
Explain Phase 📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Sample Spaces and Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. Learners are guided through a whole-class discussion to formalise the definition of 'probability space' (sample space) and the formula for theoretical probability. They then apply these definitions to three contextualised problems in their books: (1) Kamau adds a third stop — Kitengela town — to his model (T or E). Learners list the full sample space of 8 outcomes for three stops. (2) A mama mboga at Gikomba market draws one item at random from her basket containing 3 tomatoes, 2 onions, and 1 avocado — all items are distinguishable. List the sample space and find the probability of drawing an onion. (3) A student at Machakos Boys rolls a fair die once. Find the probability of rolling a prime number. Learners write	Open the Explain phase by pointing to the board predictions from Phase 1. Say: 'Look at what we predicted. Now look at what we found. Who wants to explain WHY some of us missed outcomes?' Give 10 seconds of wait time. Cold-call 2 learners. Then formally write on the board: 'SAMPLE SPACE (Probability Space) = the complete set of ALL possible outcomes of a random experiment. We write it using curly brackets { }.' Underline 'complete' and 'all.' Say: 'Theoretical Probability of event A = (number of outcomes in A) ÷ (total outcomes in the sample space). Write this in your formula box now.' For problem 1, say: 'Work silently — I want to see YOUR list, not your partner's. 4 minutes.' After 4	Formalisation after exploration (productive struggle then consolidation). Learners build definitions from their own table-filling experience rather than receiving definitions first. The three application problems escalate in abstraction (sequential stops → market items → single die) so learners see that the sample space method is general, not limited to the matatu context. The thumbs up/sideways check is a low-stakes whole-class formative signal.	Collect a 'ticket out' written response: 'Write the sample space for flipping a coin and rolling a die at the same time. How many outcomes? What is the probability of getting Heads AND a 6?' Learners write this in the last 4 minutes of the Explain phase on a half-sheet. Teacher reviews during the DQB phase to identify learners needing support before Lesson 4.

	solutions independently (8 min) then compare with a partner (3 min).	minutes, cold-call 1 learner to write their list on the board. Ask class: 'Any outcome missing? Any repeated? Thumbs up if complete, thumbs sideways if unsure.' Address gaps. For problems 2 and 3, give 4 minutes combined, then 3 minutes partner comparison. Cold-call 1 learner per problem. For problem 2, specifically ask: 'How many total outcomes are in the sample space? How many are favourable? So what is the theoretical probability?' Emphasise the link: 'This is how Kamau can say something mathematical about his chances even though he can't be certain.' Write on the board: 'Experimental probability (from tallies) ≈ Theoretical probability (from sample space) — but they are rarely identical.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Projectile Motion] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Sample Spaces and Events (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class DQB (a large chart or ARES digital board displaying Kamau's question and all accumulated lesson questions). Each learner receives one sticky note (or types into the ARES DQB tool). They write ONE thing they now understand better about Kamau's puzzle AND one new question that today's lesson raised for them. Sticky notes are placed on the DQB under two columns: 'We Now Know' and 'We Still Wonder.' The teacher reads three notes aloud and the class votes (thumbs up) if they share that understanding or question. The teacher then writes the new anchoring question for the DQB in red: 'How do we count all possibilities without missing any?' and connects it with an arrow to	Distribute sticky notes (or direct to ARES DQB). Say: 'You have 3 minutes — write one thing you NOW understand about Kamau's situation that you did not understand at the start of today, and one question you STILL have. No names needed.' After 3 minutes, collect/display notes. Read three notes aloud (choose one misconception, one good insight, one genuinely interesting question). For each, ask: 'Who else is thinking this? Thumbs up.' Then say: 'Here is the question that today's lesson officially adds to our board:' — write 'How do we count all possibilities without missing any?' Say: 'In Lesson 4 we will use this to explore what happens when two outcomes cannot both occur at the same	Metacognitive DQB reflection. The two-column structure (Know / Wonder) forces learners to distinguish between settled and unsettled understanding. Connecting today's question to previous DQB entries with an arrow makes the storyline progression visible and reinforces that science (and mathematics) is a cumulative, question-driven process.	Teacher reads all sticky notes after class. Tally the proportion of 'We Now Know' entries that correctly reference sample space or theoretical probability. Any note that reveals a persistent misconception (e.g., 'the sample space always has 4 outcomes') is flagged for explicit address at the start of Lesson 4.

	the previous lessons' questions ('Why can't Kamau predict with certainty?' from L1; 'What do probability values actually mean?' from L2).	time — like a seat being both Taken AND Empty. Can you think of an example from Kamau's route?' Give 10 seconds. Cold-call 1 learner. Leave the question unanswered — it seeds Lesson 4 (mutually exclusive events).		
Model Building Phase 📹 **VIDEO:** [Video: Projectile Motion] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Sample Spaces and Events (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their 'Kamau's Route Model' — a cumulative diagram started in Lesson 1 and updated each lesson. Today's addition: they draw a labelled 2×2 two-way table for Kamau's two-stop model and write the sample space {TT, TE, ET, EE} below it. They annotate one outcome (e.g., TT) with: 'Theoretical probability = 1/4 (if equally likely).' They draw a dotted arrow from the sample space box to their experimental probability tally (from Lesson 1), labelling the arrow: 'Experimental results approximate the theoretical probabilities.' Below the model, they write one sentence: 'Kamau cannot predict with certainty because ALL four outcomes in the sample space are possible — but probability tells him how likely each one is.' Pairs compare their updated models and check that both components (two-way table AND annotation) are present before the lesson ends.	Say: 'Open your Kamau model. You have 6 minutes to add today's layer — the two-way table, the sample space set, the probability annotation, and the connecting arrow to your Lesson 1 tally. I will come round to check.' Circulate; look specifically for: (1) correct labelling of rows and columns; (2) all four outcomes listed in set notation; (3) the fraction 1/4 written with a note 'if equally likely'; (4) the connecting arrow to Lesson 1 data. For learners who finish early: 'Extend your model — sketch what the sample space would look like for THREE stops. Don't solve it, just set up the table.' In the final 2 minutes, cold-call 1 pair to hold up their model and describe what the connecting arrow means. Close with: 'Kamau's notebook gave us experimental probability. Today's model shows us theoretical probability. They should be close — but why are they never exactly the same? That is the question for Lesson 4.'	Cumulative model revision (NGSS Science and Engineering Practice 2: Developing and Using Models). Each lesson adds a new layer to the same artefact, making conceptual progression tangible. The connecting arrow between the sample space and the Lesson 1 tally is the key sensemaking move — it physically represents the relationship between theoretical and experimental probability. Pair checking before the end ensures no learner leaves with an incomplete model.	Teacher performs a rapid gallery walk in the final 2 minutes, checking that every learner's model contains: a labelled 2×2 table, the set {TT, TE, ET, EE}, a probability fraction annotation, and the connecting arrow. Any model missing the connecting arrow or the set notation is flagged for a brief 1-on-1 clarification at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal:\n\n1. COMPLETENESS OF ENUMERATION: Did the majority of learners successfully fill all 36 cells of the 6×6 dice grid without prompting? If not, what systematic error did you observe — were learners skipping rows, confusing ordered and unordered pairs, or stopping early? How will you address this in Lesson 4?\n\n2. TRANSITION FROM EXPERIMENTAL TO THEORETICAL: When you pointed to the connecting arrow on the cumulative model, could learners articulate in their own words why the Lesson 1 tally data and the theoretical fraction 1/4 are related but not identical? If learners struggled, consider opening Lesson 4 with a quick comparison of experimental frequencies from the tally against the theoretical 1/4 — computing the difference explicitly.\n\n3. FAIRNESS DISCUSSION: Did the game-stall fairness question generate genuine debate, or did learners simply calculate 1/6 and stop? The social justice dimension (is a game stall at a school fair exploiting students?) should surface authentic engagement. If it did not, consider assigning a short home task: 'Find one game of chance in your community (e.g., a board game, a betting stall at a local market). List its sample space and decide if it is fair.'\n\n4. THREE-STOP EXTENSION: The question about Kamau's three-stop model (8 outcomes) was introduced but not fully resolved. Did learners who attempted the extension correctly identify $2^3 = 8$ outcomes? This is a seed for the multiplication principle, which will be formalised in the tree diagram lesson (Lesson 7). Note which learners spontaneously used a tree structure versus a three-way table — this reveals readiness.\n\n5. TICKET-OUT ACCURACY: Review the coin-and-die tickets. The correct sample space has 12 outcomes {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6} and P(H and 6) = 1/12. Record the proportion of correct responses. If fewer than 70% are correct, begin Lesson 4 with a 5-minute re-teaching of sample space construction for two-stage experiments before introducing mutually exclusive events.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners constructed a 2×2 two-way table for Kamau's two-stop (Mlolongo and Athi River) model, filling in all four ordered outcome pairs (TT, TE, ET, EE), and built a 6×6 grid listing all 36 ordered pairs for two dice at a school game stall fundi fair.
What did I learn?	The sample space (probability space) is the complete set of all possible outcomes of a random experiment; theoretical probability of an event equals the number of favourable outcomes divided by the total number of outcomes in the sample space; a two-way table guarantees exhaustive, non-repetitive enumeration of outcomes for two-stage experiments.
How does this explain the phenomenon?	Kamau cannot predict with certainty whether a seat will be filled at any given stop because all four outcomes in his sample space {TT, TE, ET, EE} are genuinely possible; however, the sample space gives him a mathematical tool to assign theoretical probabilities to each outcome, which approximate — but do not exactly match — the experimental frequencies recorded in his tally notebook.

LESSON 4 (80 minutes): Theoretical Probability and Single Events: What Should Kamau Expect?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to calculate theoretical probability of single events and compare these computed values against experimental frequencies gathered in earlier lessons, anchored in Kamau's matatu boarding scenario.
Knowledge	Learners will know that theoretical probability of an event is defined as P(E) = (number of favourable outcomes) ÷ (total number of equally likely outcomes), that theoretical probability values range from 0 to 1 inclusive, and that as the number of trials increases, experimental probability converges toward theoretical probability (Law of Large Numbers).
Skills	Learners will be able to: (i) identify the sample space of a single-event experiment; (ii) calculate the theoretical probability of specified single events; (iii) compare theoretical probability values against experimental frequencies from Lesson 1 data; (iv) use theoretical probability to make quantitative predictions about Kamau's matatu seat occupancy.
Attitudes	Learners will appreciate that mathematics provides a rigorous, fair, and culturally relevant tool for reasoning under uncertainty, and will value accuracy and integrity when recording and comparing probability values.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	In Lesson 3 learners mapped all possible outcomes onto sample space grids. Now, in Lesson 4, they move from listing outcomes to assigning numerical probability values to specific single events. They revisit Kamau's tally notebook data from Lesson 1 and ask: if 10 out of 14 seats are filled on an average trip, what does theory say the probability of a randomly chosen seat being occupied is — and how close did our Lesson 1 experiments come to that value? This lesson is the bridge between experimental evidence and theoretical prediction, directly advancing the driving question.
Safety Notes	No physical hazards anticipated. Remind learners to handle dice and coins carefully and not to throw them across desks. Ensure all learners have equal access to manipulatives; redistribute dice sets equitably before the activity begins.

B. LESSON OVERVIEW

Learners open by predicting whether Kamau's tally notebook data (20 trips, recorded passenger counts) will match a clean theoretical fraction. They then observe a structured dual activity — computing theoretical probabilities from sample space grids constructed in Lesson 3 (dice-sum and seat-status scenarios) and extracting relative frequencies from Kamau's Lesson 1 tally data. In the Explain phase learners compare theoretical and experimental values side-by-side on a whole-class convergence chart, articulate the Law of Large Numbers informally, and calculate the theoretical probability that a randomly chosen seat on Kamau's matatu is occupied. The DQB is updated with the new consensus question about whether theory and experiment ever perfectly coincide. Learners close by revising their cumulative probability model to include the theoretical probability formula and a convergence arrow linking experimental to theoretical values.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a projected display of Kamau's tally notebook:	Display Kamau's tally notebook table on the board or printed	Predict-before-instruction (Elicit Prior Knowledge): by committing a	Observable indicator: every learner has a written fraction and a

📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	20 trip records with passenger counts (6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14). Each learner silently reads the data for 60 seconds, then writes individual responses in their exercise books to two prompt questions: (1) 'Without calculating anything, predict: what single fraction do you think best describes the chance that a randomly chosen seat on any one trip is occupied?' (2) 'Will your predicted fraction match a fraction calculated purely from the definition of probability — or will it be different? Why?' Learners then share predictions with a shoulder partner for 2 minutes before selected pairs share with the class.	handout. Say verbatim: 'Take 60 seconds — silent, individual thinking. Write your predicted fraction and your reason in your exercise book. Do not discuss yet.' Set a visible 60-second timer. After 60 seconds say: 'Now turn and share your prediction with your shoulder partner. You have 2 minutes.' After partner talk, cold-call 4 pairs (aim for at least 1 pair who predicted 10/14, 1 who predicted a different fraction, and 1 who said they were unsure). For each response ask: 'What reasoning led you to that prediction?' Record all predicted fractions on the board in a 'Prediction Bank' column without affirming or correcting. Close with: 'Hold your prediction — by the end of today you will know whether theory agrees with you.'	prediction to paper before any instruction, learners activate and expose their prior understanding of ratio and frequency, creating a cognitive hook that makes the subsequent comparison meaningful. Recording all predictions publicly without judgment establishes psychological safety and sets up a concrete thing to 'test' during the lesson.	written reason in their exercise book within the first 3 minutes. Teacher circulates and scans books during partner talk — look for (a) correct use of 'favourable outcomes over total outcomes' language (shows readiness), (b) confusion between 10 out of 14 seats per trip versus 14 out of 20 full trips (common misconception to address in Explain phase), and (c) blank or 'I don't know' entries (these learners need targeted support during observe phase).
Observe Phase 📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of 3–4 at their desks. Each group receives: (A) their Lesson 3 sample space grid showing outcomes for two dice (the game-stall scenario from the Machakos Bus Station), and (B) a printed 'Convergence Worksheet' containing Kamau's 20-trip tally table alongside two calculation tasks. Task 1 (using the sample space grid): List all outcomes where the sum of two dice equals 7. Count favourable outcomes. Count total equally likely outcomes in the grid. Write the theoretical probability P(sum = 7). Then count how many times sum = 7 appeared in the Lesson 1 dice-rolling experiment results (posted on the board from Lesson 1). Write the experimental frequency as a fraction of total rolls. Record both values in the comparison table on the	Distribute the Convergence Worksheet and confirm groups have their Lesson 3 grids. Say: 'You have 18 minutes for both tasks. Task 1 uses the grid from Lesson 3 — no re-rolling needed, we are calculating from the list of outcomes. Task 2 uses Kamau's notebook. Both tasks ask you to compute a theoretical probability AND find the matching experimental value. Record both in the comparison table.' Circulate continuously. At the 8-minute mark do a mid-activity check: cold-call one group on Task 1 — 'How many favourable outcomes did you find for sum = 7? Tell me which squares on your grid.' (Expected: 6 outcomes: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) out of 36 total → P = 6/36 = 1/6.) Give 10 seconds WAIT TIME before accepting answers. At the 15-minute mark	Parallel-Case Comparison (Two-Context Anchoring): learners compute theoretical probability in two parallel contexts simultaneously — the abstract dice game and the concrete Kamau scenario. Seeing the same mathematical structure (favourable ÷ total) operate identically in both contexts deepens transfer. Pairing each theoretical value with its experimental counterpart from Lesson 1 on the same worksheet makes the convergence relationship visible before it is named.	Observable indicators: (1) In Task 1, groups correctly identify 6 favourable outcomes for sum = 7 on the 6×6 grid and write P = 6/36 — groups writing fewer than 6 outcomes have missed rotational pairs (e.g., wrote (1,6) but not (6,1)); teacher intervenes with: 'Is (1,6) the same outcome as (6,1) on your grid? Show me on the grid.' (2) In Task 2, the key indicator is whether learners write P(occupied) = 10/14 (correct) versus 14/20 (conflating trip frequency with seat probability) — teacher notes which groups make this error for targeted correction in Explain phase. (3) All groups produce a completed comparison table with both theoretical and experimental values side by side.

	worksheet. Task 2 (Kamau's matatu seat): The matatu has 14 seats. On an average trip, 10 seats are filled. Assume each of the 14 seats is equally likely to be the 'randomly chosen' seat. What is the theoretical probability P(chosen seat is occupied)? Write it as a fraction, a decimal, and a percentage. Compare this to the experimental relative frequency of full-seat trips from Lesson 1 tally data.	prompt groups still on Task 1: 'Move to Task 2 now even if Task 1 is not fully written up — we will consolidate Task 1 together.' For Task 2, listen for the misconception of using 14/20 (trips where all seats were full) instead of 10/14 (seats occupied per trip) — do not correct yet; flag it for the Explain phase. In the last 2 minutes ask each group to write one sentence: 'Our theoretical value and our experimental value are [close / far apart] because…'		
Explain Phase 📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class consolidation. The teacher builds a Convergence Chart on the board — a two-column table headed 'Theoretical Probability (calculated)' and 'Experimental Relative Frequency (from Lesson 1)' with rows for: (i) P(sum of two dice = 7), (ii) P(a randomly chosen seat on Kamau's matatu is occupied). Group reporters contribute their values. The class discusses: why are the two columns not identical? What would happen if Kamau ran 1,000 trips instead of 20? Learners then individually write the formal definition of theoretical probability in their own words, annotate it with the Kamau seat example, and compute two additional single-event probabilities from a mini-scenario printed at the bottom of the worksheet: 'At the Kericho tea estate canteen, a worker randomly picks one mandazi from a tray of 5 plain, 3 sugar-coated, and 2 coconut mandazi. Find: P(plain), P(not coconut), P(sugar-coated or coconut).' These three calculations are solved individually and then verified with a partner.	Build the Convergence Chart by inviting group reporters in sequence. After both rows are filled, ask the whole class: 'Look at the two columns. Are they equal?' (Wait 10 seconds — cold-call 3 learners.) 'Why not?' (Wait 10 seconds — cold-call 3 different learners.) Write on the board: 'As the number of trials INCREASES, experimental probability gets CLOSER to theoretical probability.' Say: 'This is called the Law of Large Numbers — and it is exactly why Kamau's 20-trip notebook is useful but not perfect.' Now address the 10/14 vs 14/20 misconception directly — without naming the group, write both fractions on the board and ask: 'Someone calculated 14/20 for Kamau's seat probability. Talk with your partner: what does 14/20 actually mean in Kamau's context? What does 10/14 mean?' (Wait 10 seconds — cold-call 2 pairs.) Confirm: '14/20 tells us the experimental frequency of completely full trips; 10/14 tells us the theoretical probability that a specific randomly chosen seat is occupied, based on average occupancy — these answer	Convergence Chart + Targeted Misconception Confrontation: the public side-by-side chart makes the relationship between theoretical and experimental probability concrete and discussable. The deliberate, non-punitive public examination of the 10/14 vs 14/20 misconception uses cognitive conflict to deepen conceptual clarity. The mandazi extension task transfers the theoretical probability formula to a novel Kenyan context, checking for genuine generalisation rather than rote application.	Observable indicators: (1) Every learner writes a correct formal definition of theoretical probability including both 'favourable outcomes' and 'total equally likely outcomes' — teacher collects 6 randomly chosen books after this phase to verify. (2) For the mandazi task, all three probabilities are: P(plain) = 5/10 = 1/2; P(not coconut) = 8/10 = 4/5; P(sugar-coated or coconut) = 5/10 = 1/2 — learners who add numerators without checking total (e.g., write 3+2=5 but over an incorrect denominator) are flagged for follow-up. (3) At least 80% of the class can articulate verbally why 10/14 and 14/20 answer different questions about Kamau's scenario.

		different questions.' For the mandazi task: give 5 minutes of individual silent work, then say: 'Check your answer for P(plain) with your partner.' Cold-call 2 pairs for each of the three probabilities. For each, ask: 'Did your partner get the same answer? If different, whose reasoning is correct and why?' Emphasise that P(not coconut) = 1 − P(coconut) = 1 − 2/10 = 8/10 — the complement rule emerges naturally here. Close the phase by returning to predictions: 'Look at your Prediction Bank on the board. Who had 10/14? Who had something different? What does the definition of theoretical probability tell us about why 10/14 is the mathematically justified answer?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Projectile Motion Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write on a sticky note (or a designated section of their exercise book if physical sticky notes are unavailable) their response to the prompt: 'Based on today's lesson, what do you now understand about why Kamau's tally notebook approximates but never exactly matches theory? Write one thing you NOW understand and one new question you still have.' Learners post their sticky notes on the class DQB under the column 'Lesson 4 — Theoretical Probability'. The class then votes (show of hands) on one new consensus question to add to the DQB: the teacher reads out 4–5 representative questions gathered from the sticky notes and the class votes. The leading candidate is: 'Are theoretical and experimental probabilities ever exactly the same — and if so, when?' This is added formally to the DQB in the	Distribute sticky notes or direct learners to the DQB section of their books. Say: 'You have 3 minutes — silent individual writing. One thing you NOW understand about Kamau's situation that you did not understand at the start of today's lesson. One question you still have.' After 3 minutes, invite learners to place/read their notes. Quickly scan and read aloud 4–5 questions — without attribution — that represent the class's emerging thinking. Say: 'I am going to read four questions. Raise your hand for the one that you most want us to answer before this unit ends.' Tally votes visibly. Add the winning question to the DQB. Then point to the existing DQB questions from Lessons 1–3 and ask: 'Which of our earlier questions has today's lesson helped us answer?' Cold-call 2 learners. Draw an arrow or tick on the DQB to mark progress. Close with: 'Kamau now has a	DQB as Living Knowledge Tracker: the Driving Question Board functions as a public, cumulative record of the class's growing understanding. Revisiting earlier questions and marking them as 'answered' or 'partially answered' reinforces the storyline continuity. The voting ritual gives every learner agency in shaping the inquiry direction and ensures the DQB reflects genuine student thinking rather than teacher-imposed questions.	Observable indicators: (1) Every learner produces at least one 'now understand' statement and one new question — absence of a question may indicate surface-level engagement and should prompt a brief teacher check-in. (2) The quality of 'now understand' statements is diagnostic: statements like 'theoretical probability is a fraction' indicate surface understanding; statements like 'Kamau's 20 trips are not enough trials for experiment to match theory exactly' indicate deeper understanding of convergence. Teacher photographs the DQB at the end of this phase for their own formative record.

	'Questions We Still Have' section.	theoretical probability for seat occupancy. Is that enough for him to plan his route? We will keep investigating.'		
Model Building Phase 📹 **VIDEO:** [Video: Projectile Motion] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative probability model — a visual diagram started in Lesson 1 and revised each subsequent lesson. Today they add two new elements: (1) a labelled box containing the theoretical probability formula P(E) = (number of favourable outcomes) / (total number of equally likely outcomes), with Kamau's example P(seat occupied) = 10/14 written inside; (2) a double-headed convergence arrow connecting the 'Experimental Probability' box (added in Lesson 1) to the new 'Theoretical Probability' box, labelled 'As trials increase, these get closer.' Learners also annotate the sample space section of their model (from Lesson 3) with a callout: 'Theoretical probability counts favourable cells in the sample space.' In the final 2 minutes learners share their revised model with a partner and describe one change they made and why.	Say: 'Take out your cumulative model. Today we are adding two things — the theoretical probability formula box and the convergence arrow. You have 8 minutes.' Circulate and look for: (a) models where the formula is written correctly with both components labelled, (b) models where the convergence arrow is missing or pointing the wrong direction, (c) models where the Kamau example is written with incorrect values. For (b) prompt: 'Your arrow connects experiment to theory — which direction does convergence go? Does experiment start far and get closer to theory, or the other way around?' After 8 minutes: 'Share with your partner — tell them one thing you added and why it belongs in the model.' Cold-call 2 pairs to share with the whole class. Close with: 'Your model is growing. Next lesson we will add mutually exclusive events — and ask whether Kamau can have a seat that is both occupied AND empty at the same time.'	Cumulative Model Revision (Progressive Refinement): the cumulative model serves as a personalised knowledge artefact that grows across the 8-lesson sequence. Each revision forces learners to integrate new concepts with prior ones rather than treating each lesson as isolated. The convergence arrow is a particularly powerful model element because it captures a dynamic relationship (not just a static definition), pushing learners beyond recall toward relational understanding.	Observable indicators: (1) The theoretical probability formula box is present and correctly written with both components labelled — teacher checks 6 models during circulation. (2) The convergence arrow is present, correctly directed (experimental → theoretical as trials increase), and labelled — absence or misdirection reveals a conceptual gap about the Law of Large Numbers. (3) The Kamau example within the formula box reads P(seat occupied) = 10/14 (not 14/20 or another erroneous value) — residual presence of 14/20 signals the misconception from Phase 2 was not resolved and requires individual follow-up before Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your professional journal: (1) How many learners held the 10/14 vs 14/20 misconception during the Observe phase? Was your mid-activity cold-call sufficient to surface it, or did it only emerge during whole-class consolidation? Consider whether starting Lesson 5 with a 2-minute 'misconception revisit' is warranted. (2) Did the Convergence Chart successfully make the Law of Large Numbers intuitive, or did learners treat it as a fact to memorise? If the latter, consider adding a quick simulation at the start of Lesson 5 — rolling a single die 10 times vs 50 times and comparing experimental to theoretical P(even number) — to make convergence viscerally felt. (3) Were all groups able to connect the mandazi extension task back to Kamau's scenario, or did they treat it as an unrelated calculation exercise? If the latter, strengthen the explicit transfer prompt in future iterations: 'How is choosing a mandazi from the tray structurally the same as Kamau choosing a random seat?' (4) Review the DQB questions collected. Do they suggest learners are ready for mutually exclusive events (Lesson 5), or is there unresolved confusion about sample space or experimental probability that needs consolidation? Adjust the opening of Lesson 5 accordingly. (5) Note which learners produced the highest-quality cumulative model revisions — these learners could serve as 'model narrators' in Lesson 5, briefly presenting their model to a small

group before new content is introduced.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners computed theoretical probabilities from Lesson 3 sample space grids (P(sum of two dice = 7) = 6/36) and extracted experimental relative frequencies from Kamau's 20-trip tally notebook, recording both values side-by-side in a Convergence Worksheet. They also observed the teacher-facilitated public examination of the 10/14 vs 14/20 misconception on the Convergence Chart.
What did I learn?	Learners learned that theoretical probability is defined as P(E) = (number of favourable outcomes) ÷ (total number of equally likely outcomes); that theoretical probability values range between 0 and 1; that Kamau's theoretical probability of a randomly chosen seat being occupied is 10/14 ≈ 0.71; and that as the number of experimental trials increases, experimental probability converges toward theoretical probability (Law of Large Numbers, stated informally).
How does this explain the phenomenon?	Learners explained in writing and discussion why Kamau's 20-trip experimental frequency does not exactly match the theoretical probability 10/14, connecting this to the insufficiency of small sample sizes; why 10/14 and 14/20 answer fundamentally different questions about Kamau's scenario; and why the sample space grid from Lesson 3 is the tool that makes theoretical probability countable and exact. They updated their DQB with the consensus question: 'Are theoretical and experimental probabilities ever exactly the same — and if so, when?'

LESSON 5 (80 minutes): Mutually Exclusive Events and the Addition Law	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to identify mutually exclusive events, derive the Addition Law P(A or B) = P(A) + P(B), and apply it to real-life situations including Kamau's matatu data.
Knowledge	Learners will know that mutually exclusive events cannot occur at the same time in a single trial, and that the probability of either of two mutually exclusive events occurring is found by adding their individual probabilities.
Skills	Learners will be able to: (i) identify whether two events are mutually exclusive by examining the sample space; (ii) derive and state the Addition Law for mutually exclusive events; (iii) apply P(A or B) = P(A) + P(B) to prize-draw, matatu, coin, and dice scenarios.
Attitudes	Learners will appreciate that mathematical language — 'mutually exclusive' — precisely captures everyday situations where only one outcome at a time is possible, making probability a fairer and more reliable decision-making tool.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	Lesson 5 introduces mutually exclusive events through the Kisumu school prize-draw context and connects directly back to Kamau's tally notebook: learners use the Addition Law to calculate the probability that a matatu trip is either fully booked (14 passengers) OR nearly empty (fewer than 7 passengers) — two outcomes that cannot happen at the same time. This advances the driving question by giving Kamau a precise rule for combining probabilities of distinct outcomes.
Safety Notes	No physical hazards. Remind learners that dice and coins are shared resources — handle them carefully and return them after the activity. Ensure learners are seated safely during group work rearrangement.

B. LESSON OVERVIEW

Learners begin by predicting whether two prize-draw outcomes can happen simultaneously, exposing their intuitive (often incorrect) understanding of combined probability. They then investigate a Kisumu school prize-draw sample space in pairs, discovering that mutually exclusive events share no outcomes and that their probabilities simply add. Through structured sensemaking, learners derive the Addition Law, verify it with Kamau's 20-trip tally data and a dice experiment, and apply it to new contexts. The DQB is updated with the precise definition of mutually exclusive events. The lesson closes with learners revising their cumulative probability model to include the Addition Law node.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12	Learners are shown the following scenario on the board (or printed card): 'At Kisumu Day School's end-of-year prize-giving, 30 student names are placed in a single box. One name is drawn.	Display the prize-draw scenario. Say: 'Read carefully — I want your honest first thought, not the correct answer. You have 2 minutes alone, then 1 minute with your partner.' Set a visible timer. After 3	Think-Pair-Share: Activates prior knowledge about simultaneous outcomes and creates a cognitive anchor. By recording all predictions publicly, the teacher creates accountability — learners	Observable indicator: Can the learner give a reason (even an incorrect one) for their YES/NO answer? Listen for use of phrases like 'at the same time', 'only one winner', or 'two prizes'.

🔍 Search ARES for similar videos 📖 **HTML:** Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Prize A is a geometry set; Prize B is a Kenyan atlas. Can the same draw give ONE student BOTH prizes at the same time? Write YES or NO in your exercise book and explain your reasoning in one sentence.' Learners work individually for 2 minutes, then share with a shoulder partner for 1 minute. Three pairs are cold-called to share their reasoning with the class. The teacher records all three responses on the board without evaluating them, labelling them Prediction 1, Prediction 2, Prediction 3.	minutes, cold-call exactly 3 pairs using the class register (rows 1, 3, 5). Record each answer verbatim on the board. Say: 'We are NOT judging these yet — we will come back to them at the end of Phase 2.' Point to Kamau's tally notebook displayed at the front and say: 'Keep Kamau in mind — can his matatu be both fully booked AND nearly empty on the same trip?' Allow 10-second wait time after each cold-call before accepting the response. Do not confirm or deny any prediction.	are motivated to test their own claim during the Observe phase. Connecting the prize-draw to Kamau immediately frames the new concept within the anchoring phenomenon.	Misconceptions to flag: learners who say YES because 'one student could win both if their name is drawn' — these learners have not yet grasped that a SINGLE draw can only yield ONE outcome. Note these learners for targeted questioning in Phase 2.
Observe Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Activity A — Prize-Draw Sample Space Investigation (25 minutes): Learners work in pairs. Each pair receives a printed 'Kisumu Prize-Draw Worksheet' (or teacher-drawn table on the board). The worksheet shows: 30 students numbered S1–S30. Event A = {student is in Form 3} = {S1–S12}, so 12 students. Event B = {student is in Form 4} = {S13–S30}, so 18 students. Task 1: List all outcomes in Event A. List all outcomes in Event B. Do any students appear in BOTH lists? Task 2: Calculate P(A), P(B), and P(A or B) by counting favourable outcomes over 30. Task 3: Check — does P(A) + P(B) = P(A or B)? Write a conjecture. Activity B — Kamau's Tally Check (10 minutes): Learners return to Kamau's 20-trip data displayed on the board: [6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14]. Task 4: Let Event F = trip is fully booked (14 passengers). Let Event L = trip has fewer than 7 passengers (i.e., 6 passengers). Count how many trips are F. Count how many trips are L. Can a trip be both F and L	Distribute worksheets (or direct learners to draw the table). Say: 'Task 1 is about listing — no calculating yet. Take 5 minutes.' Circulate; after 5 minutes say: 'Now Task 2 — use your sample space list, not a formula.' After Task 2, pause the class: 'Hands up if P(A) + P(B) equals P(A or B) for your pair.' Count hands aloud. Say: 'Interesting — now write ONE sentence explaining WHY you think this works or does not work. 10 seconds to think first.' Cold-call 2 pairs who did NOT raise their hands. For Activity B, say: 'Go back to Kamau's notebook. Kamau needs to know — what is the chance his next trip is either packed OR nearly empty? Can those two things happen on the same trip?' Allow 10-second wait time. Cold-call 3 learners from different rows. After Activity B, return to the board predictions from Phase 1 and ask: 'Which prediction was closest? Why?' Cross out or annotate each prediction publicly.	Data-Based Conjecture: Learners construct the Addition Law empirically before it is stated formally. By working through two contrasting contexts (prize draw and matatu tally), learners see the pattern holds across different scenarios. The explicit 'Can these happen at the same time?' question targets the core conceptual criterion for mutual exclusivity. Revisiting Phase 1 predictions closes the predict-observe loop and reinforces metacognitive awareness.	Observable indicators: (1) Learner correctly identifies that no student appears in both Event A and Event B lists (zero overlap). (2) Learner correctly counts 8 fully-booked trips and 2 trips with fewer than 7 passengers in Kamau's data, giving P(F) = 8/20 = 2/5 and P(L) = 2/20 = 1/10, P(F or L) = 10/20 = 1/2. (3) Learner verifies 2/5 + 1/10 = 4/10 + 1/10 = 5/10 = 1/2. Misconception to watch: learners who attempt to subtract overlapping outcomes — they have confused this with non-mutually exclusive events. Flag these learners for the Explain phase discussion.

	at the same time? Calculate P(F), P(L), and P(F or L) from the tally. Check: does P(F) + P(L) = P(F or L)?			
Explain Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class concept consolidation (15 minutes): The teacher formally introduces vocabulary and the law using the board. Learners copy the definition, law, and worked examples into their exercise books. Definition: Two events A and B are mutually exclusive if they cannot occur at the same time in a single trial — i.e., they share no outcomes in the sample space ($A \cap B = \varnothing$). Addition Law for mutually exclusive events: P(A or B) = P(A) + P(B). Learners then complete two application problems independently (5 minutes): Problem 1 (Dice): A fair die is rolled once. Event X = score is a multiple of 3 = {3, 6}. Event Y = score is an odd number less than 3 = {1}. Are X and Y mutually exclusive? Find P(X or Y). Problem 2 (Mandazi stall): At a Nairobi market stall, 20 mandazi are packed in a box: 8 are plain, 7 are coconut-flavoured, and 5 are cardamom-flavoured. One mandazi is picked at random. Event P = plain, Event C = coconut. Are P and C mutually exclusive? Find P(P or C). After 5 minutes, learners compare answers with the person beside them, then one volunteer per problem writes their solution on the board for class review.	Write the definition on the board slowly, underlining 'cannot occur at the same time' and 'share no outcomes'. Say: 'The key test is always: can these two things happen in the SAME single trial? If yes — not mutually exclusive. If no — mutually exclusive.' Write P(A or B) = P(A) + P(B) in a box. Point to Kamau's result on the board: 'We already proved this with Kamau's data. The law just names what we discovered.' For independent problems, say: 'Work alone for 5 minutes — I want to see YOUR thinking, not your neighbour's.' Circulate and identify one strong solution and one with an error for each problem. After pair-check, cold-call the volunteer for Problem 1 and ask the class: 'Do you agree? What would change if X and Y shared an outcome?' Allow 10-second wait time. For Problem 2, ask: 'Can a mandazi be both plain AND coconut at the same time?' to reinforce the physical meaning of mutual exclusivity. Address the flagged misconception from Phase 2 explicitly: 'Some of us started subtracting — that is for a different situation we will meet later. For now, if events cannot overlap, we simply ADD.'	Explicit Concept Formalisation followed by Guided Practice: The teacher names and defines what learners discovered empirically, creating a bridge from inductive observation to deductive application. The use of a familiar Kenyan food context (mandazi) for Problem 2 lowers cognitive load for the new law, letting learners focus on the concept rather than decoding an unfamiliar scenario. Public board work by volunteers externalises thinking and creates a shared reference.	Observable indicators: (1) Learner correctly identifies $X \cap Y = \varnothing$ for the dice problem and calculates P(X or Y) = 2/6 + 1/6 = 3/6 = 1/2. (2) Learner correctly identifies P and C as mutually exclusive (a mandazi cannot be both plain and coconut) and calculates P(P or C) = 8/20 + 7/20 = 15/20 = 3/4. (3) Learner can articulate WHY the events are mutually exclusive in words, not just by formula. Red flag: learner who writes P(P or C) = (8+7)/20 correctly but cannot explain why — prompt with 'Why didn't you subtract anything?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Mutually	The class DQB (a large poster or section of the board maintained across all 8 lessons) is updated collaboratively (7 minutes). The teacher displays the current DQB with entries from Lessons 1–4.	Point to the DQB and say: 'This board is our collective thinking over 5 lessons. Today we add something specific.' Give learners 2 minutes to write their contributions. Say: 'Your We-Now-	Collaborative Knowledge Mapping via DQB: The DQB externalises the growing model of understanding across the unit, making the storyline visible. Requiring the phrase 'mutually	Observable indicators: (1) Learner's We-Now-Know statement correctly uses 'mutually exclusive' and references the Addition Law or the 'cannot happen at the same time' criterion.

Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are asked to: (1) Write one new 'We now know' statement on a sticky note or in their books to add to the DQB. (2) Write one new 'We still wonder' question that today's lesson raised. The teacher reads selected contributions aloud and physically places them on the DQB under the columns 'What we now know' and 'What we still wonder'. The teacher then reads the updated driving question aloud: 'How can Kamau — and other Kenyans facing uncertain everyday outcomes — use the mathematics of probability to make smarter, fairer, and safer decisions?' and asks: 'How does today's Addition Law move Kamau closer to a useful answer?' One learner is cold-called to answer.	Know must include the phrase mutually exclusive. Your We-Still-Wonder must be a genuine question — not something we already answered today.' Read 4–5 sticky notes aloud before placing them. For the driving question connection, say: 'Kamau now knows P(fully booked OR nearly empty) = 1/2. What does that mean for his planning?' Allow 10-second wait time before cold-calling. Write the DQB update prompt on the board: 'What makes two events mutually exclusive?' and circle it as today's DQB headline question. Ensure the DQB visually shows progression: Lessons 1–4 entries lead logically to today's Addition Law entry.	exclusive' in the We-Now-Know statement reinforces precise mathematical vocabulary. The driving question connection forces learners to evaluate the utility of the new law — not just its mechanics — which deepens conceptual understanding.	(2) Learner's We-Still-Wonder question is genuinely forward-looking (e.g., 'What if events CAN overlap — do we still add?', 'Does the Addition Law work for three events?'). (3) Cold-called learner can connect P(F or L) = 1/2 to a practical implication for Kamau (e.g., 'Half his trips are either very full or very empty — he can plan for both extremes').
Model Building Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Cumulative Model Revision (8 minutes): Learners open their probability model diagram begun in Lesson 1 and updated each subsequent lesson. Today they add a new node labelled 'Mutually Exclusive Events & Addition Law' connected to the existing nodes for 'Experimental Probability', 'Sample Space', 'Theoretical Probability', and 'Range of Probability'. In the new node, learners must write: (a) the definition of mutually exclusive events in their own words; (b) the Addition Law formula P(A or B) = P(A) + P(B); (c) one example from today's lesson (prize draw, Kamau's tally, dice, or mandazi stall); (d) an arrow labelled 'NEXT: What if events CAN overlap?' pointing to a blank node for Lesson 6. Learners share their updated model with a partner and check each other's definition for accuracy. One pair is cold-called to read their definition aloud.	Say: 'Your model grows every lesson — today it gains its most important node yet. You have 6 minutes to add all four elements.' Circulate and check that definitions include 'cannot happen at the same time' and 'no shared outcomes'. Prompt struggling learners: 'Look at your prize-draw result — can S1 be in Form 3 AND Form 4 at the same time? Write that idea in your own words.' After 6 minutes, say: 'Share with your partner — read your definition out loud and let them tell you if it is complete.' Cold-call one pair and ask the class: 'Give a thumbs-up if their definition captures the idea correctly.' Address any incomplete definitions immediately. Close by pointing to Kamau's notebook display: 'By Lesson 8, Kamau's model will be complete enough to help him make a real staffing decision. Today's law is a crucial piece of that model.'	Cumulative Visual Modelling: The growing diagram makes the knowledge structure of the unit tangible and personal. Requiring a forward-pointing arrow ('NEXT: What if events CAN overlap?') primes learners for Lesson 6 on non-mutually exclusive events and the general Addition Rule, creating productive anticipatory thinking. Peer-checking of definitions builds accountability and mathematical communication skills.	Observable indicators: (1) Learner's model node includes all four required elements. (2) Definition correctly captures both criteria: 'cannot happen at the same time' AND 'no shared outcomes in the sample space'. (3) Example is correctly labelled as mutually exclusive (not just any probability example). (4) Arrow to next node is present, indicating the learner anticipates a more complex case. Exit-ticket equivalent: collect 5 randomly selected model diagrams at the end of class to check for completeness before Lesson 6.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did the majority of learners correctly identify the zero-overlap criterion as the defining feature of mutually exclusive events, or did they focus only on the formula? If learners applied P(A or B) = P(A) + P(B) mechanically without being able to justify WHY, plan a 5-minute re-teach at the start of Lesson 6 using a counter-example (non-mutually exclusive events) to sharpen the distinction. (2) In Activity B, did learners correctly extract P(F) = 8/20 and P(L) = 2/20 from Kamau's 20-trip tally? If several pairs obtained wrong counts, the issue may be in reading frequency tables from Lessons 1–2 — address this as a prerequisite gap before Lesson 6 introduces tree diagrams. (3) Were the We-Still-Wonder entries on the DQB genuinely anticipatory of non-mutually exclusive events, or did they repeat content already covered? If the latter, learners may not yet perceive the logical gap that Lesson 6 will fill — spend 2 extra minutes at the start of Lesson 6 posing a deliberately overlapping event scenario (e.g., 'What if Event A = trip has more than 10 passengers AND Event B = trip has an even number of passengers?') to create productive confusion. (4) Was the Kisumu prize-draw context culturally accessible and motivating? Note any learner comments for future context refinement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners investigated a Kisumu Day School prize-draw sample space and Kamau's 20-trip matatu tally to identify events that share no outcomes.
What did I learn?	Two events are mutually exclusive when they cannot occur at the same time in a single trial (they share no sample space outcomes), and the probability of either occurring is found using the Addition Law: P(A or B) = P(A) + P(B).
How does this explain the phenomenon?	Learners explained why P(fully booked OR fewer than 7 passengers) = P(F) + P(L) = 8/20 + 2/20 = 1/2 for Kamau's matatu, and why the formula works only because a trip cannot be both fully booked and nearly empty at the same time.

LESSON 6 (80 minutes): Independent Events and the Multiplication Law: Can Kamau Count on Both Trips?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate whether the outcome of one random event affects a second event, derive P(A and B) = P(A) × P(B) for independent events, and apply the multiplication law to predict the probability that Kamau's matatu is full on both the morning and afternoon trip.
Knowledge	Learners will know that: (i) two events are independent when the occurrence of one does not change the probability of the other; (ii) the multiplication law states P(A and B) = P(A) × P(B) for independent events; (iii) dependent events arise when outcomes are affected by prior selections (e.g., drawing passengers without replacement); (iv) tree diagrams display all outcomes of two successive independent events and allow direct reading of combined probabilities.
Skills	Learners will be able to: (i) design and conduct a coin-toss experiment to test for independence; (ii) construct a sample space and a tree diagram for two successive independent events; (iii) apply P(A and B) = P(A) × P(B) to solve real-life probability problems; (iv) distinguish between independent and dependent events using everyday Kenyan contexts; (v) update the Driving Question Board with evidence-based claims.
Attitudes	Learners will appreciate that mathematical reasoning about independence helps everyday decision-makers — like Kamau and Kericho tea farmers — plan more fairly and confidently despite uncertainty, and will value honest, systematic data collection as the foundation for reliable probability statements.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	In Lessons 1–5 learners built experimental probability, identified sample spaces, distinguished mutually exclusive events, and used tree diagrams for sequential events. Lesson 6 now asks the deeper question: does the morning trip's outcome change what we should expect in the afternoon? By proving independence through experiment and sample-space analysis, learners unlock the multiplication law — the most powerful tool yet for helping Kamau plan staffing and fuel budgets across multiple trips.
Safety Notes	No hazardous materials are used. Coins should be clean Kenyan shillings. Learners must not flip coins near their eyes. If physical coins are unavailable, the teacher may use a random-number table or a spinner made from cardboard. Ensure desks are cleared to avoid coins rolling onto the floor and causing slipping hazards.

B. LESSON OVERVIEW

Lesson 6 of 8 in the evidence-gathering sequence for Sub-Strand 3.2: Probability I. Learners open by predicting whether the result of a first coin toss affects the second, then conduct a structured 40-toss paired experiment using Kenyan 5-shilling coins to collect frequency data. They build a full sample space grid and a tree diagram, observe that each branch probability remains unchanged regardless of what happened on the first toss, and formally derive P(A and B) = P(A) × P(B). A Kericho tea-farmer scenario (will it rain on both Monday and Tuesday?) bridges the abstraction to local context before the anchor phenomenon is revisited: What is the probability that Kamau's matatu is full on BOTH the morning and afternoon trip, given historical data from his tally notebook showing full trips (14 passengers) occurred 8 out of 20 times? Learners compute P(full morning AND full afternoon) = 0.4 × 0.4 = 0.16, then contrast with a dependent-event preview (drawing two passengers from a queue without replacement) to sharpen the independence concept. The Driving Question Board is updated with a new claim card: "Two events are independent when knowing the result of one gives you no new information about the other — and then you can multiply their probabilities."

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Independent, Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a Kenyan 5-shilling coin and the following prompt card (teacher-distributed): 'Kamau tosses a coin to decide which driver takes the morning shift. The coin lands HEADS. He tosses again for the afternoon shift. Does the first result change what is likely to happen on the second toss? Write your prediction and your reasoning in your exercise book. Use the sentence starters: I predict… because… I am (certain / fairly sure / unsure) because…' Learners write independently for 3 minutes, then share predictions with a shoulder partner. Four volunteers share whole-class. Teacher records all predicted positions on the chalkboard under two columns: 'YES — first toss affects second' and 'NO — each toss is fresh'. Learners also predict: 'If Kamau's matatu has a 2-in-5 chance of being full on any single trip, what is the probability it is full on BOTH the morning AND afternoon trip?' — they write a numeric guess before any instruction.	Distribute coin and prompt card simultaneously to save time. Say: 'Before we do any mathematics today, I want your honest gut feeling. No wrong answers at this stage — your prediction is evidence of your thinking.' Allow 3 minutes of silent writing. Use WAIT TIME of 12 seconds after asking 'Who predicted YES, that the first toss affects the second?' before cold-calling. Cold-call exactly 4 learners (2 from each column) using lolly-stick names. Record every prediction verbatim on the board — do NOT correct or affirm. Ask: 'Kamau's notebook shows 8 full trips out of 20. Without any formula, what number would YOU guess for the chance of two full trips in a row?' Cold-call 3 more learners. Leave all predictions visible throughout the lesson so learners can revisit them. Pose the anchoring question: 'By the end of this lesson you will be able to tell Kamau — with mathematics — exactly how to calculate that chance. Let's find out whether your gut was right.'	Elicitation of prior knowledge via prediction writing: making learners commit a written position before instruction creates cognitive dissonance when evidence contradicts the prediction, deepening the eventual sensemaking. Recording predictions publicly holds learners accountable and gives the teacher immediate formative data about misconceptions (most learners will predict dependence — the 'gambler's fallacy').	Observable indicator: every learner has a written prediction with reasoning in their book within 3 minutes. Teacher scans books during partner share, noting learners who leave the reasoning blank (these need scaffolding in Phase 2). Chalkboard tally of YES/NO predictions gives the teacher a class-level misconception map — if >70% say YES, plan an extra cold-call in Phase 3 specifically targeting the gambler's fallacy.
Observe Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Independent,	EXPERIMENT — 'The Kericho Tea-Farmer's Double Coin Toss': Learners work in pairs. Each pair tosses their Kenyan 5-shilling coin TWICE per trial and records the ordered outcome (e.g., Heads-Heads, Heads-Tails, Tails-Heads, Tails-Tails) in a tally table with four rows. Each pair conducts 20 trials (40 individual tosses). After 20 trials, pairs combine data with the neighbouring pair (groups of 4) to get a 40-trial dataset, then the	Before pairs begin, model ONE trial: 'Watch me. I toss once — I get Heads. I record H in the first-toss column. I toss again — I get Tails. I record T in the second-toss column. My outcome for this trial is H-T. I put one tally mark in the H-T row.' Use a large tally table drawn on the board as a reference. Say: 'Your job is to be rigorous — like Kamau with his tally notebook. Every toss counts.' Circulate during the experiment; after 5	Data-driven pattern recognition: by pooling class data into ~160 trials, experimental frequencies converge on ¼ for each of the four outcomes, making it visually obvious that the second toss is not skewed by the first. This large-n pooling strategy mirrors scientific practice (replication increases reliability) and directly echoes how Kamau's 20-trip notebook gives better estimates than a single day's observation. The Kericho	Observable indicators: (1) Each pair has a completed 20-row tally table with four outcome rows summing to 20. (2) The 2×2 sample space grid shows all four outcomes correctly. (3) Experimental probabilities are calculated as fractions and decimals. Teacher checks 4–6 books during circulation; specifically looks for learners who have written unequal theoretical probabilities (e.g., P(HH) = ½) — a

Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	whole class pools all results on a teacher-drawn master tally on the board (targeting ~160 trials total for a class of 40). Alongside the experiment, learners complete a structured worksheet with two tasks: (A) Draw the full sample space grid (a 2×2 table: first toss on one axis, second toss on the other) and list all four equally likely outcomes. (B) From their pair's 20 trials, calculate the experimental probability of each of the four outcomes and compare to the theoretical probability of ¼ each. Learners also read a two-paragraph teacher-created scenario card: 'Amina is a tea farmer in Kericho. The local agricultural officer tells her there is a 3-in-5 chance of rain on any given day in October. Amina wants to know the chance it rains on BOTH Monday AND Tuesday so she can plan when to harvest. She assumes rain on Monday does not change the chance of rain on Tuesday — the two days are independent.' Learners underline the key phrase 'does not change' and discuss in pairs why this assumption might be reasonable.	minutes ask the class: 'Raise your hand if, after 10 trials, you are seeing one outcome much more than the others.' Use this as a real-time check. After pairs finish, rapidly collect whole-class totals: call on one representative from each group of 4 to read their four frequencies aloud; teacher tallies on the board. Once master tally is complete, ask: 'Look at the total frequency for HH compared to HT — are they roughly equal? What does that suggest about whether the first toss is controlling the second?' Allow WAIT TIME of 15 seconds. Cold-call 3 learners. Direct learners to the Amina/Kericho scenario card: 'Read this in 2 minutes. When you find the phrase that explains independence, circle it and tell your partner what it means in your own words.'	scenario card introduces independence linguistically ('does not change') before the formal definition, anchoring the abstract concept to a locally recognisable agricultural decision.	sign of residual dependence thinking requiring targeted intervention in Phase 3.
Explain Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Independent, Conditional, and Mutually Exclusive Events	PART A — Deriving the Multiplication Law: The teacher guides a whole-class discussion using the pooled tally data and the sample space grid. Learners are directed to calculate: 'From your 20 trials, how many times did you get Heads on the first toss regardless of the second? Write that as a probability.' (Expected: ≈ ½.) 'Of all the trials where the first toss was Heads, what fraction also showed Heads on the second toss?' (Expected: ≈ ½ of the first-Heads trials, i.e., P(HH) ≈ ¼.)	For Part A, write the column-fraction calculation on the board step by step, narrating: 'I am asking — of all my Heads-first trials, how often did I also get Heads second? If that fraction equals my overall probability of Heads, the first toss gave me NO extra information about the second.' Pause 12 seconds. Ask: 'Does this match or contradict your prediction from Phase 1?' Cold-call 4 learners specifically from the 'YES — first affects second' prediction column on the board,	Conceptual bridging through contrast (CER — Claim, Evidence, Reasoning): learners first use their own experimental data as evidence, then formalise the multiplication law as a claim, then reason through two contrasting scenarios (independent coin tosses / Kamau's trips vs. dependent queue selection). The contrast makes the boundary condition of independence crisp and memorable. The tree diagram serves as a dual-purpose tool — both a calculation device and a	Observable indicators: (1) Every learner's book shows the calculation P(full morning AND full afternoon) = 0.4 × 0.4 = 0.16 with correct working. (2) The Kericho tree diagram has all four end-branch probabilities correctly labelled and summing to 1. (3) The one-sentence contrast (independent vs. dependent) is written in every book. Teacher uses a show-of-hands check: 'Thumbs up if you can tell me in one word what MUST stay the same for two events to be

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners see numerically that P(second = H | first = H) ≈ P(second = H) — the conditional probability equals the unconditional probability — so the events are independent. Teacher writes on the board the formal rule: 'For independent events A and B: P(A and B) = P(A) × P(B).' Learners copy this into their notes with a coloured box around it. PART B — Kericho Application: Learners calculate, individually: 'P(rain Monday) = 3/5. P(rain Tuesday) = 3/5. If the two days are independent, what is P(rain on BOTH days)?' Answer: 3/5 × 3/5 = 9/25 = 0.36. They then draw a tree diagram with two branches at each level (Rain / No Rain) and label all four end-branch probabilities, verifying that the four branch-end probabilities sum to 1. PART C — Kamau's Matatu (Anchor Phenomenon): Learners use Kamau's tally notebook. Full trips (14 passengers) occurred 8 times out of 20. So P(full trip) = 8/20 = 2/5 = 0.4. Teacher poses: 'Are the morning trip and the afternoon trip independent? Think about whether Kamau's morning passengers deciding to board changes what afternoon passengers will do.' Learners discuss in pairs (2 minutes), then whole-class: teacher cold-calls 3 learners. Consensus: yes, independent (different sets of passengers make independent decisions). Therefore: P(full morning AND full afternoon) = 0.4 × 0.4 = 0.16. Learners interpret: 'Kamau can expect both trips to be full about 16 times in every 100 days — roughly once a week.' PART D — Dependent Event Preview: Teacher poses the contrast: 'Imagine 14 passengers	inviting them to revise or defend their view. For Part B (Kericho), say: 'Work individually for 4 minutes. Calculate P(rain both days) and draw the full tree diagram. I will cold-call three people to explain their tree, not just give the answer.' Circulate and place a small tick on books that correctly label all four end-branches. For Part C, explicitly connect to the phenomenon: 'Kamau has been staring at his notebook for months. Today he has a formula. What does 0.16 tell him that he could NOT see just by reading his tally?' Allow WAIT TIME of 15 seconds; cold-call 2 learners. For Part D, signal the boundary clearly: 'This dependent case is where we are heading in Probability II. For now, your job is to recognise the difference — I want you to be able to explain in one sentence why the matatu example is independent and the queue example is dependent.' Give 2 minutes for learners to write that sentence.	visual representation of how probabilities propagate without change across branches when events are independent.	independent' — target word is 'probability'. If fewer than 75% of thumbs are up, re-teach Part A using a second example before moving to Phase 4.

	are waiting at the Kitengela stage. Kamau's matatu can take all 14. He randomly selects the FIRST passenger to sit in the front seat: 5 of the 14 are women. Then he selects a SECOND passenger for the other front seat. Are these two selections independent?' Learners discuss in pairs. Teacher draws the tree diagram showing P(first = woman) = 5/14, then P(second = woman | first = woman) = 4/13 ≠ 5/14 — the probability CHANGED, so these events are DEPENDENT. Teacher labels this clearly: 'This is a preview of Probability II — drawing without replacement creates dependence. Today we focus on independent events.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Plinko Probability] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Independent, Conditional, and Mutually Exclusive Events (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner receives two sticky-note-sized card slips (or tears a page in two). On CARD 1 (green — a new CLAIM): they write a response to the DQB anchor question 'How do we know when two events are truly independent?' using the sentence frame: 'Two events are independent when… We can test this by… An example from Kamau's route is…' On CARD 2 (orange — a remaining QUESTION): they write one thing they are still unsure about or a new question that arose from the dependent-event preview, e.g., 'How do we calculate P(A and B) when events are NOT independent?' or 'Does independence mean the events can never happen at the same time?' Learners post their green CLAIM cards in the 'What We Now Know' section of the class DQB and their orange QUESTION cards in the 'What We Are Still Wondering' section. The teacher reads 4–5 green cards aloud and	Give learners 5 minutes to write both cards independently — no partner discussion during writing, to ensure individual accountability. Circulate and quickly scan for cards that conflate independence with mutual exclusivity (a common confusion — a learner might write 'independent means they cannot both happen at the same time', which describes mutual exclusivity, not independence). If you see this, do NOT correct during writing; instead, select that card for reading aloud and use the class response (thumbs) to surface the misconception whole-class. When reading green cards aloud, ask: 'Class — thumbs up if this claim is complete, sideways if it needs a small fix, down if something is wrong.' After thumbs, say: 'Who can add one word or phrase to make this claim stronger?' Cold-call 2 learners. Close by pointing to the full DQB and saying: 'Compare today's green cards with the claims from Lesson 5. Notice	Metacognitive consolidation through public knowledge display: the DQB functions as a living class model of growing understanding. By physically posting claims and questions, learners externalise their thinking, creating a shared artefact that tracks the storyline's progression. Distinguishing green (claim) from orange (question) cards reinforces the scientific practice of separating what is known from what remains uncertain — directly mirroring Kamau's situation of knowing probabilities but not certainties.	Observable indicators: (1) Every learner posts exactly one green and one orange card. (2) Green cards reference the multiplication law or the concept of independence explicitly — vague cards ('probability is useful') are flagged for follow-up. (3) Orange cards reveal whether learners are ready for dependent events (Lesson 7) — a card asking 'what happens without replacement?' signals readiness; a card asking 'what is probability?' signals a learner needing foundational review. Teacher photographs the DQB at the end of Phase 4 to track conceptual progression across the unit.

	invites the class to agree, qualify, or challenge each claim using a simple thumbs-up/sideways/down response. The teacher also reads 3 orange question cards aloud and tells learners which will be addressed in Lessons 7–8 (laws of probability, combined events, probability tree diagrams with more complex scenarios).	how our understanding of Kamau's problem is getting more precise. We started asking WHY he cannot predict — now we can calculate exact probabilities for specific combinations of events.'		
Model Building Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Independent, Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Probability Model' page — a single annotated diagram they have been building across all 8 lessons. Today they add two new elements: (1) A TREE DIAGRAM section: Learners draw a clean two-stage tree diagram for Kamau's morning and afternoon trips, labelling branches P(Full) = 2/5 and P(Not Full) = 3/5 at each stage, and calculating all four end-branch probabilities: P(F and F) = 4/25, P(F and NF) = 6/25, P(NF and F) = 6/25, P(NF and NF) = 9/25. They verify these sum to 25/25 = 1. (2) A RULE BOX: Next to the tree diagram they write and box the multiplication law: 'Independent Events: P(A and B) = P(A) × P(B)', and below it the contrast statement: 'Dependent Events: P(A and B) ≠ P(A) × P(B) — must use conditional probability (Probability II).' (3) An INTERPRETATION STATEMENT: 'This means Kamau can expect both trips to be full on about 16 out of every 100 operating days — roughly once a working week. This helps him plan fuel budgets for high-revenue days.' Learners who finish early extend the model by adding a third trip branch (if three consecutive trips were independent, what is P(all three full)?) and computing $(2/5)^3$ =	Say: 'You have been building this model since Lesson 1. Today it gets its most powerful upgrade yet — a formula that lets you combine probabilities across multiple events. Open your model page and let's add two things.' Display a teacher model of the completed tree diagram on the board as a reference. After 8 minutes of individual drawing, ask: 'Pause. Check that your four end-branch probabilities add to exactly 1. If they don't, find the arithmetic error.' Allow 2 minutes for self-correction. Circulate and specifically check that learners label the branches P(Full) = 2/5, not P(Full) = 8 — they must write probabilities, not raw frequencies. For the interpretation statement, say: 'Kamau is not a mathematician — write your statement so he could read it and immediately use it to plan his week. No symbols — plain sentences.' Cold-call 2 learners to read their interpretation aloud. Close the lesson by asking: 'Hold up your model page. Look at how much it has grown since Lesson 1. Next lesson, we will add the final law — what happens when we want EITHER event A OR event B to occur — and we will discover that addition and multiplication laws together give Kamau a	Cumulative model revision (progressive diagrammatic refinement): the running model page functions as a personal knowledge artefact that makes conceptual growth visible to both learner and teacher. Adding the tree diagram today alongside the multiplication law rule box creates a dual representation — symbolic and graphical — reinforcing the same mathematical relationship through two modalities. The interpretation statement requirement forces learners to translate formal mathematics into actionable language, building the real-world application competency mandated by the KICD.	Observable indicators: (1) Tree diagram has correct branch labels (fractions, not counts) and four end-branch probabilities summing to 1. (2) Multiplication law is written in the rule box using correct notation. (3) Interpretation statement is written in plain language and references a specific, plausible real-world implication for Kamau. Teacher reviews 6–8 collected model pages after the lesson and provides written feedback focusing on: branch labelling accuracy, arithmetic in end-branch products, and quality of plain-language interpretation. Learners who correctly compute the extension task (three consecutive trips) are noted for advanced grouping in Lesson 7.

	8/125 = 0.064.	complete probability toolkit.' Collect 6–8 model pages for written feedback (rotate whose work is collected each lesson).		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions and record brief notes: (1) MISCONCEPTION CHECK — How many learners initially predicted that the first coin toss affects the second (gambler's fallacy)? Did the pooled experimental data successfully shift their thinking, or do several learners still conflate independence with mutual exclusivity? If the misconception persisted in more than 25% of DQB green cards, plan a 5-minute re-teach at the start of Lesson 7 using the contrast: 'Mutually exclusive = cannot happen together at the same time; Independent = one happening does not change the probability of the other — they CAN happen together.' (2) EXPERIMENT QUALITY — Did pairs complete 20 trials each within the allocated time? If not, reduce to 15 trials next time or prepare a pre-printed tally sheet to reduce set-up time. (3) KAMAU CONNECTION — Did learners produce meaningful plain-language interpretation statements linking 0.16 to Kamau's planning? If statements were too abstract ('P = 0.16 means probability'), use the start of Lesson 7 to model one strong interpretation and one weak one side-by-side and ask learners to vote and explain. (4) DEPENDENT EVENTS PREVIEW — Did the queue-without-replacement example create confusion or productive curiosity? If learners seemed lost, reassure them that dependent events are the focus of Probability II and today's goal was only to recognise the difference. (5) DIFFERENTIATION — Were fast finishers engaged with the three-consecutive-trips extension? Prepare an additional challenge for Lesson 7: 'If Kamau adds a third trip to Kitengela, what is the probability that at least ONE of the three trips is full?' (requiring the complement rule — connecting forward to Lesson 7).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners conducted a paired coin-toss experiment (20 trials per pair, ~160 trials pooled class-wide) using Kenyan 5-shilling coins, recording outcomes in a four-row tally table and building a 2×2 sample space grid. They observed that the frequency of each of the four ordered outcomes (HH, HT, TH, TT) was approximately equal (~¼ each), and that the conditional frequency of Heads on the second toss given Heads on the first toss was approximately equal to the unconditional frequency of Heads — providing empirical evidence that the two tosses are independent.
What did I learn?	Learners formalised the definition of independent events — two events are independent when the occurrence of one does not alter the probability of the other — and derived the multiplication law: P(A and B) = P(A) × P(B). They applied this law in two Kenyan contexts: (i) Amina the Kericho tea farmer calculating P(rain on both Monday and Tuesday) = 3/5 × 3/5 = 9/25 = 0.36, supported by a fully labelled two-stage tree diagram; and (ii) Kamau's matatu route, calculating P(full morning AND full afternoon) = 2/5 × 2/5 = 4/25 = 0.16, interpreted as 'roughly once a working week'. Learners also previewed dependent events through the queue-without-replacement contrast, distinguishing P(second = woman | first = woman) = 4/13 ≠ 5/14 from the independent case.
How does this explain the phenomenon?	Learners explained, through their cumulative model page and DQB claim cards, that two events are independent when knowing the result of one gives no new information about the probability of the other, and that this independence allows probabilities to be combined simply by multiplication. They articulated — in plain language — what the probability of 0.16 means for Kamau's operational planning, and distinguished this independent scenario from the dependent queue-selection preview. The DQB was updated with the new claim: 'Two events are independent when the probability of the second is the same whether or not the first occurred — and then P(A and B) = P(A) × P(B) gives us the exact combined chance.'

LESSON 7 (80 minutes): Probability Tree Diagrams: Mapping All of Kamau's Possible Outcomes	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct and interpret probability tree diagrams for two sequential independent events, multiply probabilities along branches, and add probabilities across branches to find the probability of compound outcomes — applied to Kamau's matatu boarding scenario and a road-safety pedestrian context.
Knowledge	Learners will know that: (i) a tree diagram displays all possible outcomes of sequential events on labelled branches; (ii) probabilities along any path are multiplied (multiplication rule for independent events); (iii) probabilities of mutually exclusive outcomes satisfying a condition are added (addition rule); (iv) all terminal-branch probabilities in a complete tree sum to 1.
Skills	Learners will be able to: (i) construct a two-stage probability tree diagram with correctly labelled branches and probabilities; (ii) multiply along branches to compute the probability of each combined outcome; (iii) add across relevant branches to answer conditional questions; (iv) verify that all terminal probabilities sum to 1; (v) apply tree diagrams to a road-safety civic scenario.
Attitudes	Learners will appreciate that: (i) tree diagrams make hidden probability structure visible and checkable; (ii) systematic enumeration reduces errors compared to informal reasoning; (iii) probability tools carry civic responsibility — the road-safety tree raises awareness of pedestrian danger at undesignated crossing points in Nairobi.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	This is Lesson 7 of 8 in the evidence-gathering sequence. Earlier lessons established experimental probability (Lessons 1–2), sample space (Lesson 3), mutually exclusive events (Lesson 4), independent events (Lessons 5–6). Now learners synthesise all these ideas into a single visual tool — the tree diagram — that lets Kamau model two consecutive boarding stops at once, see every possible outcome, and calculate exact probabilities. The DQB question 'How do tree diagrams help us see all possible outcomes at once?' is formally answered here and recorded on the board.
Safety Notes	No physical hazards. The road-safety tree diagram discusses pedestrian near-miss statistics; teacher should handle this sensitively — some learners may have personal experience of road accidents. Frame discussion around civic awareness and data, not personal tragedy.

B. LESSON OVERVIEW

In this 80-minute lesson learners move from knowing that two boarding events are independent (Lesson 6) to representing them visually and computing compound probabilities using a tree diagram. The lesson opens by returning to Kamau's tally notebook: from the 20-trip data set, the experimental probability that a given seat is filled at Stop 1 is estimated as 14/20 = 7/10, and that it is filled at Stop 2 (independently) is also 7/10. Learners first predict what fraction of trips would have both seats filled, using only their intuition. They then construct a full two-stage tree diagram — labelling four terminal branches (Filled–Filled, Filled–Empty, Empty–Filled, Empty–Empty) — and discover the multiplication rule by verifying that the four branch products sum to 1. A second tree diagram models a road-safety scenario: the probability that a pedestrian in Nairobi's CBD crosses at an undesignated point is 3/5, and given that they cross there, the probability of a near-miss is 2/5; learners use the tree to find P(undesignated crossing AND near-miss). The DQB is updated with a new anchor card: "A tree diagram forces every possible outcome into the open — and the multiplication rule gives each outcome its exact probability." Lesson connects to KICD SLOs (e), (f) and (g): laws of probability, tree diagrams for independent events, and real-life application including road safety PCI.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown Kamau's tally notebook data on the board (the 20-trip passenger counts). The teacher poses: 'From the data, we estimated earlier that the probability a seat is filled at any stop is 7/10. Kamau's matatu now stops at TWO places: Mlolongo then Athi River. Assuming a passenger's decision at Stop 1 has nothing to do with what happens at Stop 2 — predict in your exercise book: what is the probability that BOTH seats are filled? Write ONE number. Do not discuss yet.' Learners write individual predictions silently for 2 minutes. Teacher then asks learners to hold up fingers: 1 finger = 'more than half', 2 fingers = 'exactly half', 3 fingers = 'less than half'. A quick show of hands reveals the spread of intuition. Teacher cold-calls 3 learners to share their numeric predictions and reasoning, recording them on the board without correcting. Teacher asks: 'Could a picture of all the possibilities help us check these predictions? That is exactly what we will build today.'	1. Display the 20-trip tally table and the previously established P(Filled) = 7/10 from Lesson 6. Say verbatim: 'We know each stop is independent — like two separate coin flips. Your job right now is to PREDICT, not calculate. What does your gut say?' 2. Allow 2 minutes silent individual prediction. Do NOT allow pair talk during this phase. 3. After 2 minutes, say: 'Show me fingers — more than half, exactly half, less than half.' Scan the room. 4. WAIT TIME: after calling on each of 3 cold-called learners, wait 10 seconds before prompting. Record their predictions verbatim on the board (e.g., 'Amina says 1/2, Brian says 49/100, Cynthia says 7/10'). 5. Do not affirm or correct. Say: 'By the end of this lesson, you will know exactly who is closest — and WHY the correct answer is not obvious without a diagram.'	Prediction Anchoring: by committing to a numeric prediction before instruction, learners activate prior knowledge of independence and create a personal stake in the answer. This creates productive cognitive dissonance when the tree diagram reveals the multiplicative structure, making the multiplication rule memorable rather than arbitrary.	Observable indicator: every learner has written a numeric prediction in their exercise book. Teacher scans 5–6 books during the finger-show. Red flag: if more than half the class predicts exactly 7/10 (i.e., they think two-event probability equals one-event probability), teacher notes this as the key misconception to address explicitly in Phase 3.
Observe Phase 📹 **VIDEO:** Mutually Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and	PART A — Building the Tree (25 min): Learners receive a structured worksheet (or draw in exercise books following teacher board-work). Step 1: Teacher draws the trunk — a dot labelled 'Start (Kamau leaves Machakos Country Bus Station)'. Step 2: Two first-level branches grow rightward — 'Stop 1: Filled (F_1)' with probability 7/10 written above the branch, and 'Stop 1: Empty (E_1)' with probability	1. Draw tree diagram slowly on board in stages — do NOT draw the full diagram at once. At each stage, PAUSE and ask a checking question before continuing. 2. At Step 2: 'Why do 7/10 and 3/10 add to 1?' — cold-call 1 learner, wait 10 seconds. Expected answer: because filled and empty are mutually exclusive and exhaustive. If wrong, redirect: 'Look at your Lesson 4 notes on mutually	Staged Diagram Construction (Gradual Reveal): the tree is built in four discrete steps with a checking question between each step. This prevents learners from copying a completed diagram without understanding its structure. The 'sum-to-1 verification' is a built-in self-assessment tool that makes errors visible immediately, reinforcing the mathematical structure rather than depending on	Observable indicators: (i) every learner's tree has four terminal branches with products summing to 1 — teacher checks 6–8 books during circulation; (ii) learners can state verbally WHY Stop 2 probabilities are identical on both sides (independence); (iii) the road-safety terminal probability 6/25 is correctly computed. Red flag: learners who add probabilities along a path (obtaining 7/10 + 7/10

Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	3/10 written below. Teacher asks: 'Why must these two branch probabilities add to 1?' (Cold-call 1 learner, 10-second wait.) Step 3: From each first-level node, two second-level branches — 'Stop 2: Filled (F_2)' and 'Stop 2: Empty (E_2)' — each labelled 7/10 and 3/10 respectively. Teacher says: 'These probabilities are the SAME on both sides — why? Discuss with a partner for 60 seconds.' Step 4: Learners label four terminal nodes and write the PRODUCT along each path: F_1F_2 = 7/10 × 7/10 = 49/100; F_1E_2 = 7/10 × 3/10 = 21/100; E_1F_2 = 3/10 × 7/10 = 21/100; E_1E_2 = 3/10 × 3/10 = 9/100. Step 5: Learners ADD all four terminal values: 49+21+21+9 = 100/100 = 1. Teacher: 'This is your CHECK — if your terminal probabilities do not sum to 1, you have made an error.' PART B — Road-Safety Tree (10 min): Teacher introduces a new scenario on the board: 'The Nairobi City County traffic data shows that the probability a randomly chosen pedestrian crosses at an undesignated point in the CBD is 3/5. Given they cross at an undesignated point, the probability they are involved in a near-miss is 2/5. Given they cross at a designated crossing, the probability of a near-miss is 1/10.' Learners draw a second two-stage tree in their books. Branches: Cross type (Undesignated 3/5, Designated 2/5) → Near-miss outcome (Near-miss / No near-miss). They label all four terminal branches and compute the probability of 'Undesignated AND Near-miss' = 3/5 × 2/5 = 6/25.	exclusive events.' 3. At Step 3: 'Why are the Stop 2 probabilities the same on both branches?' — pair-share 60 seconds then cold-call 1 learner. Expected answer: independence — Stop 2 boarding does not depend on what happened at Stop 1. 4. At Step 4: circulate and look at 4–5 books while learners compute products. If a learner adds instead of multiplies, ask quietly: 'Are these events happening one AFTER the other, or are these alternatives?' 5. After Step 5 verification: 'Raise your hand if your four products sum to exactly 1.' If hands are not universal, say: 'Compare with your neighbour — find the arithmetic error before we move on.' 6. For PART B road-safety tree: say verbatim: 'This tree is not just mathematics — it carries a message. What does a probability of 6/25 for an undesignated crossing near-miss tell us about road safety in Nairobi?' Allow 30-second think time before cold-calling 2 learners.	teacher validation.	= 14/10, which exceeds 1) have confused 'along a path' with 'across outcomes' — teacher addresses this in Phase 3.
Explain Phase	PART A — Formalising the Rules	1. Write 'MULTIPLICATION RULE'	Rule Extraction from Evidence:	Observable indicators: (i) learners

📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	(10 min): Teacher leads a whole-class discussion to draw out two explicit rules from what learners observed: (1) Multiplication Rule along a branch: 'To find the probability of a sequence of independent events, MULTIPLY the probabilities along the path.' (2) Addition Rule across branches: 'To find the probability that ANY ONE of several mutually exclusive outcomes occurs, ADD the terminal probabilities.' Learners write both rules in their own words in a 'Rules' box in their exercise book. Teacher then returns to the predictions from Phase 1 written on the board. The class confirms that P(both seats filled) = 49/100 — checking which prediction was closest. PART B — Extended Application (10 min): Pairs are given a challenge: 'Using Kamau's tree, find: (a) P(at least one seat is filled across the two stops); (b) P(exactly one seat is filled).' Learners must identify which terminal branches satisfy each condition and add those branches. For (a): $F_1F_2 + F_1E_2 + E_1F_2 = 49/100 + 21/100 + 21/100 = 91/100$. For (b): $F_1E_2 + E_1F_2 = 21/100 + 21/100 = 42/100$. Pairs share answers with an adjacent pair (Think-Pair-Square). PART C — Civic Connection (5 min): Teacher facilitates brief discussion: 'The road-safety tree shows P(undesignated AND near-miss) = 6/25. What advice would you give the Nairobi City County based on this tree diagram? Is 6/25 high or low compared to P(designated AND near-miss) = 2/5 × 1/10 = 2/50 = 1/25?' Learners compare 6/25 versus 1/25 and articulate that crossing at an undesignated point is 6 times	and 'ADDITION RULE' as headings on the board. Ask: 'In your own words, when do we multiply and when do we add on a tree diagram?' Cold-call 3 learners — one for multiply, one for add, one to synthesise. Wait 10 seconds after each question. 2. Return to Phase 1 predictions on the board. Say verbatim: 'Let's honour Amina's prediction — she said [X]. The tree gives us 49/100. Why might our intuition fail here?' Draw out the idea that multiplication of fractions less than 1 always gives a SMALLER number. 3. For the extended application: circulate during pair work. Listen for: do learners correctly identify which branches to add for 'at least one'? If a pair includes E_1E_2 in their 'at least one' calculation, ask: 'Read back the terminal label on that branch — does it include any seat being filled?' 4. For Part C civic discussion: say verbatim: 'A mathematician does not just compute — they communicate findings to help people. What would you put on a road-safety poster using these numbers?' Cold-call 2 learners. Validate responses that reference the 6× difference.	rather than presenting the multiplication and addition rules as given formulas, the teacher draws them out of what learners have already observed in Phase 2. This inductive approach (NGSS Science and Engineering Practice 6 — Constructing Explanations) means learners own the rules as conclusions of their own investigation. The civic connection uses SEP 7 (Engaging in Argument from Evidence) by asking learners to form a data-based recommendation.	can state both rules correctly in their own words in their exercise books; (ii) P(at least one seat filled) = 91/100 and P(exactly one seat filled) = 42/100 are correctly derived by both members of at least 4 pairs spot-checked during circulation; (iii) learners can articulate the 6× risk comparison for the road-safety scenario. Red flag: any learner who still adds along a path after Phase 3 instruction requires a one-on-one redirect with a physical analogy ('If you flip two coins, both getting heads is LESS likely than one flip getting heads — so you multiply, which makes the number smaller, not larger.').

	more dangerous in terms of near-misses.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Mutually Exclusive Events Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Theoretical and Experimental Probability (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher draws attention to the DQB, which has been accumulating cards since Lesson 1. The current open question from Lesson 6 reads: 'How do we calculate the probability of two independent events happening together?' The teacher asks learners to write a NEW card answering this question using today's language — in 15 words or fewer. Sample expected response: 'Multiply along branches of a tree diagram to get the probability of both events.' Three learners share their cards aloud; the class votes (thumbs up/sideways/down) on clarity. The selected card is pinned to the DQB under the heading 'ANSWERED — Lesson 7'. The teacher then poses the NEW open question for Lesson 8: 'Now that we have experimental probability, sample space, mutually exclusive events, independent events, and tree diagrams — how do we bring all of these tools together to help Kamau make a final decision about his matatu route?' This card is pinned as 'OPEN'. Teacher also highlights the journey card: 'Lesson 1 — Kamau's tally was just numbers. Today — it is a tree with exact probabilities on every branch.' Learners spend 2 minutes in their exercise books writing a personal 'DQB journal entry': one thing that is now clear, one thing still uncertain.	1. Point to the existing DQB cards from Lessons 1–6. Say verbatim: 'Look at how far we have come — from Kamau's puzzling tally notebook to a complete tree diagram with four calculated outcomes. What question from Lesson 6 can we now fully answer?' 2. Allow 90 seconds silent card-writing. Do not suggest wording. 3. Call on 3 learners to read their cards. After each, ask the class for a thumbs vote on clarity. If a card is unclear, ask the author: 'Can you replace one word to make it sharper?' 4. Pin the winning card. Say: 'This card now represents what the whole class knows. Kamau knows it too.' 5. Pose the Lesson 8 open question aloud and pin the new card. 6. Allow 2 minutes for personal DQB journal entries — circulate and read 3–4 without commenting, noting remaining confusions for Lesson 8 opening.	Cumulative DQB Narrative: the DQB functions as a public, evolving record of the class's growing understanding of the driving question. Pinning an 'ANSWERED' card and a new 'OPEN' card in the same move shows learners that knowledge builds progressively — each lesson closes one door and opens another. The personal journal entry externalises metacognitive monitoring (NGSS SEP 8 — Obtaining, Evaluating, and Communicating Information).	Observable indicators: (i) every learner has written a DQB card that references either 'multiply', 'branches', or 'tree diagram'; (ii) the journal entry identifies at least one specific concept (not just 'I understand probability') as clear; (iii) teacher's scan of 3–4 journal entries reveals quality of remaining confusion for Lesson 8 planning.
Model Building Phase 📹 **VIDEO:** [Mutually	Learners return to the cumulative probability model they have been building since Lesson 1 — a large annotated diagram of Kamau's matatu route with probability	1. Say verbatim: 'Your model started as a simple matatu route map. Today it gets its most powerful addition — a tree. Open your model to the two-stop	Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models): the running matatu-route model is the physical record of the unit's conceptual	Observable indicators: (i) every learner's cumulative model now contains a correctly labelled two-stage tree diagram with four terminal probabilities summing to

Exclusive Events Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	notation added at each lesson. Today's addition: learners draw a miniature two-stage tree diagram in the 'Machakos Bus Station → Mlolongo → Athi River' section of their model. They label all four terminal branches with computed probabilities and add a caption: 'Tree diagram shows ALL possible boarding outcomes across two stops. Multiply along branches. Add across outcomes for conditions.' They also add a second small tree for the road-safety pedestrian scenario in the 'Nairobi CBD' section of their route diagram, labelling P(undesignated ∩ near-miss) = 6/25 with a road-safety symbol (⚠️). Finally, learners write a 'Model Update Statement' at the bottom of their diagram: 'I have added tree diagrams. My model now shows not just individual probabilities but ALL combinations of outcomes, with exact probabilities for each. Kamau can now calculate P(both stops filled) = 49/100 without guessing.' Learners share models in pairs and identify one difference between their two models, which they reconcile before the lesson ends.	section.' 2. Circulate for 8 minutes while learners draw and annotate their trees. Prioritise checking: (i) branch probabilities are correctly placed (on the branch, not at the node); (ii) terminal branch products are computed correctly; (iii) the four products are written in a small sum-check box. 3. If a learner's four products do not sum to 1: crouch to their level and say quietly: 'Your tree is telling you something is wrong — what does the sum give you?' Do not correct directly; let them self-correct. 4. After 8 minutes, prompt pair sharing: 'Compare your model with your partner. Point to ONE branch where you have different numbers. Whose is correct — and how do you know?' 5. With 2 minutes remaining, cold-call 1 learner to read their 'Model Update Statement' aloud. Ask the class: 'Does this statement accurately capture what a tree diagram adds to Kamau's knowledge?' Thumbs vote. Affirm and close.	growth. Each lesson's addition is visible and layered. Today's tree diagram addition is the most structurally rich contribution so far — it subsumes sample space (all terminal nodes), independence (identical branch probabilities on both sides), multiplication rule, and addition rule in a single diagram. The Model Update Statement requires learners to articulate the explanatory power of the new addition, not just draw it.	1; (ii) the road-safety tree with 6/25 annotation is present; (iii) the 'Model Update Statement' references Kamau and uses the word 'all' or 'every' in reference to outcomes. Red flag: models that show only one branch (e.g., only the Filled–Filled path) indicate the learner has not grasped the exhaustive nature of a tree diagram — teacher flags these learners for targeted support in Lesson 8 opening.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PREDICTION SPREAD — Did the Phase 1 prediction show the expected misconception (P(both) = 7/10)? If most learners correctly predicted a smaller value, the multiplication rule may have been partially internalised from Lesson 6 — advance Lesson 8 accordingly. (2) BRANCH DIRECTION — Did any learners draw the tree growing upward or downward instead of left-to-right? This orientation confusion can obscure the sequential logic; next time, explicitly name the left-to-right convention as 'time moves to the right — just like Kamau's journey.' (3) ADDITION CONFUSION — The most common error in tree diagram work is adding along a branch instead of multiplying. Did you encounter this? How many learners self-corrected via the sum-to-1 check versus needing teacher intervention? (4) CIVIC ENGAGEMENT — Did the road-safety scenario generate genuine discussion, or did learners treat it as a routine calculation? If engagement was low, consider opening Lesson 8 with a real Nairobi road-safety statistic from the National Transport and Safety Authority (NTSA) to ground the mathematics in lived reality. (5) DQB QUALITY — Read the 'OPEN' question cards from today's DQB journal entries tonight. Which confusions are most common? Use the top two to write the opening retrieval question for Lesson 8. (6) MODEL COMPLETENESS — How many models had all four terminal probabilities summing to 1 without teacher intervention? This is your clearest single indicator of lesson success. Target: ≥ 80% of learners. If below 60%, open Lesson 8 with a 10-minute whole-class tree diagram

re-construction from scratch using a different context (e.g., two consecutive githeri or chapati purchases at the school canteen).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a two-stage probability tree diagram built step-by-step for Kamau's matatu boarding scenario (Stop 1: Mlolongo; Stop 2: Athi River), with P(Filled) = 7/10 and P(Empty) = 3/10 at each stop. They watched all four terminal branch probabilities computed as products and verified that the four values sum to 1. They also observed a second tree diagram for a Nairobi CBD road-safety scenario showing P(undesignated crossing AND near-miss) = 6/25.
What did I learn?	Learners learned that: (1) a tree diagram displays ALL possible outcomes of sequential events simultaneously; (2) probabilities are MULTIPLIED along each branch path (multiplication rule for independent events); (3) probabilities are ADDED across branches to find the probability of any condition spanning multiple outcomes; (4) the sum-to-1 check is a built-in error detector; (5) P(both seats filled across two stops) = 49/100; P(at least one seat filled) = 91/100; P(exactly one seat filled) = 42/100; (6) crossing at an undesignated point in Nairobi is 6× more likely to result in a near-miss than crossing at a designated point.
How does this explain the phenomenon?	Learners explained that tree diagrams give Kamau a complete mathematical picture of all two-stop boarding combinations, replacing guesswork with exact probabilities. The multiplication rule works because independent events combine by scaling — each successive probability shrinks the likelihood of a specific sequence. The addition rule works because mutually exclusive terminal outcomes can be combined without overlap. The road-safety tree shows that probability is not just an academic tool — it quantifies real civic risk and can inform policy, safety campaigns, and personal decisions about where to cross the road in Nairobi.

LESSON 8 (80 minutes (double lesson)): Application, Games, and Unit Synthesis — Advising Kamau with the Full Power of Probability	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students synthesise all probability concepts from Lessons 1–7 by designing a fair probability-based game, computing and comparing experimental and theoretical probabilities from Kamau's full 20-trip dataset, writing a formal Probability Report for Kamau, completing the Driving Question Board, and constructing the final annotated probability framework diagram.
Knowledge	– Experimental probability = frequency of outcome ÷ total trials; theoretical probability = favourable outcomes ÷ total equally likely outcomes; the two converge as trials increase (Law of Large Numbers). – A fair game requires each player to have an equal theoretical probability of winning; fairness is verified by computing P(player wins) for all players and confirming equality. – The full probability framework connects: experimental probability → sample space → theoretical probability → Addition Law (mutually exclusive events) → Multiplication Law (independent events) → tree diagrams → real-life decision-making.
Skills	– Design a probability-based game specifying sample space, theoretical probabilities, and fairness conditions, and evaluate another group's game for fairness. – Compute experimental probability from a 20-trial dataset, compare it to a theoretical model, and write a structured advisory report communicating findings to a non-specialist audience.
Attitudes	– Appreciate that mathematics provides rigorous, evidence-based tools for making fair and informed decisions under uncertainty in everyday Kenyan life. – Value collaborative critique: evaluate peers' game designs constructively, recognising that honest mathematical scrutiny improves fairness and equity.
Key Inquiry Question	How is probability applied in real-life situations?
Purpose in Storyline	This is the culminating lesson: learners return to Kamau's full 20-trip notebook, apply every tool built across the unit to produce a formal Probability Report advising him on staffing decisions, and complete the DQB and final annotated model — answering the driving question definitively.
Safety Notes	No specific hazards. If physical spinners or dice are used for game design, ensure materials are handled responsibly and not thrown. Confirm all group work is conducted respectfully.

B. LESSON OVERVIEW

This double lesson is the capstone of the entire Probability I unit. It is structured in three interlocking acts that mirror the real-world process of mathematical inquiry: design, critique, and advise. In Act 1 (game design and peer evaluation), groups design a simple probability-based game — a spinning wheel, card draw, or coin-and-dice hybrid — specifying the sample space, theoretical probability for each player, and a written fairness justification. Groups then rotate and evaluate a neighbouring group's game using the probability laws from Lessons 5–7, producing written verdicts with calculations. This activity makes abstract laws concrete and personally owned: learners are not merely applying rules but defending them in front of peers.

In Act 2 (the Probability Report), the class returns to Kamau's anchoring phenomenon. Using his complete 20-trip tally (passenger counts: 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14), learners count that 8 out of 20 trips were fully booked (count = 14), giving an experimental probability of P(full trip) = 8/20 = 0.4. They then build a theoretical model: if each of the 14 seats is filled independently with probability p = 10/14 ≈ 0.714 (derived in Lesson 4), the probability that ALL 14 are filled is (10/14)^14 ≈ 0.021 — far below 0.4 — revealing that the seats are NOT filled independently (boarding is clustered: school groups, market days, word-of-mouth). This mismatch is the richest mathematical moment of the unit: it shows that real data can falsify a model, and a better model (e.g., treating the trip as a single event with P = 0.4 estimated from data) is more useful for Kamau's planning. Learners write a structured one-page Probability Report advising Kamau: on 40% of trips he should expect a full bus; he should plan to run a second vehicle roughly 2 out of every 5 morning departures; and he cannot predict any single trip with certainty but can budget staffing across a working week.

In Act 3 (DQB completion and final model), all sticky notes on the Driving Question Board are revisited. The teacher facilitates a whole-class review, marking each question as ANSWERED or REVISED. Learners then individually complete the final annotated probability framework diagram — a visual map connecting all six conceptual nodes of the unit — and share it with a partner for peer annotation. The lesson closes with a teacher-led return to the original driving question: students are cold-called to articulate, in one sentence, how Kamau — and any Kenyan facing uncertain outcomes — can use probability to make smarter, fairer, and safer decisions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a fresh copy of Kamau's 20-trip tally notebook (the original dataset from Lesson 1) and, without computing anything, write individual written predictions in their exercise books answering three prompts: (1) 'Roughly what fraction of Kamau's trips do you think ended with a full bus?'; (2) 'If you designed a perfectly fair spinning-wheel game for a school fete, what would it look like?'; (3) 'What single piece of probability advice would you give Kamau right now?' Students work silently for 3 minutes, then share predictions with a partner (Think-Pair-Share) for 2 minutes.	Display Kamau's original tally table on the board (trip numbers 1–20 with passenger counts: 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14). Say: 'We have lived with Kamau's puzzle for seven lessons. Today we answer it — with full mathematical power. But before we calculate anything, I want your best prediction right now. Look at the data. Think silently. Write.' [WAIT TIME: 3 minutes silent writing — do not prompt.] After 3 minutes: 'Turn to your partner. Compare your predictions on all three questions. Do you agree? Disagree? Note where you differ.' [WAIT TIME: 2 minutes pair discussion.] Then cold-call 3 different students on prompt 1 ('What fraction?'), 2 different students on prompt 2 ('What does a fair game look like?'), and 2 students on prompt 3 ('What advice?'). Record a range of predictions on the board in a 'Prediction Bank' column — do NOT confirm or correct yet. Say: 'Hold these predictions. By the end	Prediction Banking: recording all student predictions publicly before instruction creates cognitive commitment — students are motivated to check their own reasoning against evidence. The three-prompt structure activates prior knowledge from all seven previous lessons simultaneously, surfacing the conceptual threads the lesson will weave together.	Teacher scans written predictions during the 3-minute silent phase, noting (a) whether students recall that a full trip requires all 14 seats — indicating recall of sample space from Lesson 3; (b) whether fairness definitions invoke equal probability — indicating retention of the Addition Law context from Lesson 5; (c) whether advice references uncertainty rather than certainty — indicating internalisation of the core probability scale lesson (Lesson 2). Flag any student still predicting 'Kamau can know for certain' — they will need targeted support in the Explain Phase.

		of today, you will know exactly who was closest and why.'		
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ACT 1 — GAME DESIGN (25 minutes): Groups of 3–4 receive a 'Game Design Brief' card. Each group must design a probability-based game for a school fete at their school. The game must specify: (a) the physical setup (spinner divided into sectors, a standard deck of cards, two dice, or a bag of counters — using materials available in a Kenyan classroom); (b) the complete sample space, listed explicitly; (c) the theoretical probability of each player winning, calculated and shown; (d) a written fairness verdict — is the game fair? If not, how would they adjust it to make it fair? Groups record all work on a single A3 sheet (or two taped exercise-book pages). After 15 minutes of group work, groups rotate their A3 sheets one position clockwise. Each group then spends 8 minutes peer-evaluating the game they have received: they must identify at least one correct probability calculation, at least one error or fairness concern (if any), and write a signed 'Fairness Verdict' using a calculation to justify it. ACT 2 — PROBABILITY REPORT (25 minutes): The class reconvenes. The teacher projects Kamau's full dataset. Guided by three structured questions on the board, learners individually compute: (Q1) the experimental probability of a fully booked trip from the 20-trip data; (Q2) what a naive independent-seats theoretical model predicts for P(full trip); (Q3) why the two answers differ, and what this means for how Kamau should plan. Learners write	ACT 1 setup: Distribute Game Design Brief cards. Say: 'Your task is to design a probability game for a school fete — like the fetes at Nairobi School, Starehe Boys, or your own school. The game must be mathematically fair. You have 15 minutes. Use any of the probability tools from Lessons 1 to 7.' [WAIT TIME: allow 15 minutes of genuine group work — circulate and ask probing questions: 'How do you know your sample space is complete?', 'What does the Addition Law say about these two outcomes?', 'Is this game fair — what calculation proves it?'] After 15 minutes: 'Stop writing. Rotate your sheets clockwise. You are now a probability judge. Your job: check the maths, find any errors, and write a fairness verdict with calculations. You have 8 minutes.' [WAIT TIME: 8 minutes.] After rotation: cold-call 2 groups to share the most interesting fairness concern they found. Say: 'Good. You have just done what a professional statistician does — design, then audit.' ACT 2 setup: Project Kamau's data. Say: 'Let us now return to where we began. Here is Kamau's complete notebook.' Point to the 20 trip counts. Ask: 'How many of these trips were fully booked? Count them now.' [WAIT TIME: 30 seconds.] Cold-call 1 student. Confirm: 8 trips show count = 14. Write on board: P(full trip, experimental) = 8/20 = 0.4. Then say: 'Now here is a challenge. In Lesson 4, we said the probability that any single seat is occupied is about 10/14. If the 14 seats fill	Design-Then-Audit (Game Design + Peer Evaluation): designing a game requires learners to produce probability knowledge, while auditing a peer's game requires them to evaluate it — two different cognitive operations that together consolidate the Addition and Multiplication Laws more deeply than passive practice. Model-Data Mismatch as a Sensemaking Engine: displaying the experimental value (0.4) alongside the naive theoretical prediction (0.021) creates productive dissonance — learners must reason about WHY the model fails, which deepens understanding of independence as an assumption, not a given.	During game design, teacher checks that: (a) every group has listed a complete sample space (not just outcomes for one player); (b) probabilities sum to 1 across the sample space; (c) the fairness verdict includes a calculation, not just a claim. During report writing, teacher circulates and checks: (a) experimental probability computed correctly as 8/20 = 2/5 = 0.4; (b) student engages with the model mismatch — they must name at least one real-world reason why independence fails (clustering, market days, school groups, etc.); (c) advice to Kamau is specific and quantitative (e.g., '2 out of 5 trips', 'roughly 8 Mondays per term'), not vague ('sometimes it is full').

	a structured one-page 'Probability Report for Kamau' in their exercise books using a teacher-provided scaffold with three headings: WHAT THE DATA SHOWS / WHAT A MATHEMATICAL MODEL PREDICTS / MY ADVICE TO KAMAU.	independently, what is the theoretical probability that ALL 14 are occupied?' [WAIT TIME: 2 minutes for individual calculation.] Cold-call 2 students. Guide toward (10/14)^14 ≈ 0.021. Write both values on the board side by side: EXPERIMENTAL = 0.4, THEORETICAL (independent model) = 0.021. Say: 'These are very different. Why? Discuss with your partner for 90 seconds.' [WAIT TIME: 90 seconds.] Cold-call 3 pairs. Confirm: seats do not fill independently — market days, school trips, and social groups cause clustering. Say: 'Now write your Probability Report for Kamau. Use the three headings on the board. You have 15 minutes.'		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Two groups share their game designs and the class votes on fairness using a show-of-hands probability audit. The teacher then leads a structured 'Final Explanation' sequence: the class constructs, step by step, the complete probability framework connecting all six conceptual nodes. Learners copy and annotate the framework diagram. Three volunteers then read aloud their Probability Report advice paragraphs. The class evaluates each piece of advice against the mathematical evidence, proposing refinements. Finally, each learner writes one 'Final Explanation Sentence' answering the driving question in their own words.	Say: 'Let us hear from two groups about their game designs.' Select one group whose game is fair and one whose game has a fairness flaw. After each presentation, ask the class: 'Thumbs up if you agree this game is fair — thumbs down if you found an error. Explain your vote with a calculation.' [WAIT TIME: 15 seconds before calling on volunteers.] Cold-call 3 students for each game. After both presentations, draw the Final Probability Framework on the board as a connected diagram with six labelled nodes: [1] EXPERIMENTAL PROBABILITY (Kamau's tally) → [2] SAMPLE SPACE (all possible boarding outcomes) → [3] THEORETICAL PROBABILITY (favourable ÷ total) → [4] ADDITION LAW (mutually exclusive: P(A or B) = P(A)+P(B)) → [5] MULTIPLICATION LAW + TREE DIAGRAMS (independent: P(A and B) = P(A)×P(B)) → [6] REAL-LIFE DECISION MAKING	Connected Framework Diagram: building the six-node diagram collaboratively, with learners supplying the lesson links and Kamau examples, externalises the entire unit's conceptual structure. This is a 'knowledge mapping' strategy — it makes tacit connections explicit and visible, supporting long-term retention and transfer. Final Explanation Sentences: asking each learner to synthesise the entire unit into one precise sentence is a high-leverage sensemaking move; it forces integration of experimental evidence, theoretical tools, and practical application into a single coherent claim.	Teacher evaluates Final Explanation Sentences against three criteria: (1) Acknowledges uncertainty — uses language like 'cannot predict any single trip' or 'no certainty for one event'; (2) Invokes a specific probability tool — names experimental probability, theoretical probability, a law, or a tree diagram; (3) Connects to practical action — mentions Kamau making a decision (staffing, fuel budget, second vehicle). A strong sentence satisfies all three criteria. Sentences satisfying only criteria 1–2 indicate partial synthesis; sentences satisfying only criteria 1 indicate a student still at the descriptive level who needs follow-up. Teacher annotates 3–5 exercise books during the writing phase.

		(Kamau's staffing plan). As you draw each node, ask: 'Which lesson did we build this in? What was the key discovery?' Cold-call one student per node (6 cold-calls total). Say: 'Now copy and annotate this framework. Add one example from Kamau's story at each node. You have 5 minutes.' [WAIT TIME: 5 minutes.] Then say: 'Let us hear Kamau's advice. Three volunteers — read your advice paragraph aloud.' After each reading, ask: 'Does this advice use the experimental probability correctly? Does it acknowledge uncertainty? Is it specific enough for Kamau to act on?' [WAIT TIME: 10 seconds after each question.] Cold-call 2 evaluators per report. Close with: 'Now write one sentence — your Final Explanation Sentence — answering this question: How can Kamau use mathematics to make smarter decisions even though he can never predict any single trip with certainty?' [WAIT TIME: 2 minutes silent writing.] Cold-call 5 students to share their sentences. Write three strong sentences on the board.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability	The class gathers around or views the Driving Question Board that has been maintained since Lesson 1. All sticky notes — questions posted across eight lessons — are visible. Working in their groups, learners read through all posted questions and categorise them: FULLY ANSWERED (green dot), PARTIALLY ANSWERED (yellow dot), REVISED — the question turned out to be wrong or needed rephrasing (orange dot). Each group selects one question from the board, writes its full answer on	Say: 'Let us return to where we began — our Driving Question Board. Since Lesson 1, you have been posting questions about Kamau's puzzle and about probability. Today we answer them all.' Hand each group a sheet of three colour-coded dot stickers (green, yellow, orange). Say: 'Read every question on the board. With your group, decide: is this question fully answered, partially answered, or did we need to rephrase it? Dot each question. Then pick ONE question, write the	DQB Synthesis Review: revisiting every posted question with colour-coded resolution is a metacognitive consolidation strategy. It makes the learning trajectory visible — students literally see how their understanding grew from Lesson 1 to Lesson 8. The act of writing a full answer on a new sticky note and posting it beside the original question closes the inquiry loop, giving learners a concrete record of their own intellectual growth. This strategy also identifies	Teacher observes the quality of sticky-note answers: strong answers cite a specific lesson, name the probability concept, and include a numerical example (e.g., 'Lesson 5, Addition Law: P(6 seats OR 14 seats) = P(6) + P(14) = 2/20 + 8/20 = 10/20 = 0.5, because these outcomes are mutually exclusive'). Weak answers are vague ('we learned about probability'). Teacher photographs the final state of the board for portfolio evidence. The distribution of green vs. yellow dots informs

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	a new sticky note citing the specific lesson and probability concept that resolved it, and posts the answer note beside the question. The class then reviews the board together, and the teacher facilitates a whole-class discussion identifying which questions were answered by which lessons. Any question that remains genuinely open is discussed briefly — these become recommendations for further study.	full answer on a new sticky note — citing the lesson and the concept — and post it.' [WAIT TIME: 6 minutes group work.] After dotting: 'Let us review. Which questions got green dots? Read one aloud and give the answer.' Cold-call 4 groups, one question each. After each answer, ask: 'Which lesson answered this? What was the key mathematical idea?' For any yellow-dotted questions: 'What part is still unclear? What would we need to do to fully answer it?' For any orange-dotted questions: 'Why did we need to revise this question? What did we think at the start that turned out to be incomplete?' [WAIT TIME: 10 seconds per question type.] Finally, point to the original driving question displayed at the top of the board: 'Why is it impossible for Kamau to know for certain whether the next seat will be filled — yet we can still say something mathematical and useful about his chances?' Say: 'Does our board now answer this? Vote: thumbs up if yes.' Expect near-universal agreement. Cold-call 2 students to state the answer in their own words.	genuine remaining gaps, which the teacher can address in assessment or extension tasks.	the teacher about which concepts need reinforcement in the next unit or in formative assessments.
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Theoretical and Experimental Probability	Each learner individually completes the Final Cumulative Model in their exercise book. The model is an annotated probability framework diagram with six labelled nodes (matching the diagram built in the Explain Phase) plus Kamau's specific data at each node: Node 1 shows the experimental probability calculation (8/20 = 0.4); Node 2 shows the sample space for a single seat (filled or empty) and for the full trip; Node 3 shows the	Say: 'You have been building a model of probability since Lesson 1. Today you complete it. Open your exercise book to a fresh double page. Draw the six-node framework we built together. At each node, record Kamau's specific data or example. You have 8 minutes.' [WAIT TIME: 8 minutes individual silent work — do not offer hints; this is an individual assessment opportunity.] After 8 minutes: 'Find your Lesson 1 notes or your first	Cumulative Model Completion with L1 vs L8 Comparison: asking learners to place Lesson 1 thinking alongside their final model is a 'before-and-after' metacognitive strategy that makes conceptual growth tangible and personally meaningful. Peer annotation (green star + orange question mark) adds a low-stakes collaborative evaluation layer that surfaces remaining misconceptions without the pressure of formal assessment.	Teacher collects or photographs 6–8 completed final models during the peer-annotation phase to assess: (a) correctness of all six node labels and their connections; (b) accuracy of Kamau's data at each node (especially P = 8/20 = 0.4 at Node 1, P = 10/14 at Node 3, tree diagram at Node 5); (c) quality of the 'How my thinking changed' paragraph — does it show genuine conceptual shift, or merely surface-level vocabulary acquisition? The final models

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	theoretical single-seat probability (10/14 ≈ 0.714); Node 4 shows a mutually exclusive events example from Kamau's route; Node 5 shows the multiplication law applied to two consecutive trips and the tree diagram summary; Node 6 shows the Probability Report advice. Learners then compare their final model to a quick sketch they made in Lesson 1 (or, if unavailable, write a brief 'How my thinking changed' paragraph). They share their completed model with a partner for a 3-minute peer annotation — each partner adds one green star (something strong) and one orange question mark (something to clarify further). The lesson closes with the teacher re-reading the original phenomenon aloud and asking the class to respond in unison: 'Can we answer Kamau's question now?'	rough idea about probability. Look at what you wrote then. Write three sentences: what you thought probability was in Lesson 1, what you know now, and what surprised you most.' [WAIT TIME: 3 minutes.] Then: 'Share your completed model with your partner. Give them one green star and one orange question mark on their diagram. Be specific — write the star and question mark directly on their diagram.' [WAIT TIME: 3 minutes peer annotation.] After peer annotation: cold-call 3 pairs: 'What green star did you give? What orange question mark?' Finally, stand at the front and say: 'Let me read to you exactly what Kamau's notebook said on Day 1 of our unit.' Read aloud the original phenomenon (Kamau noticing some mornings every seat is taken, others only 6 board). Then ask: 'Can we answer Kamau's question now? Show me with your hands — thumbs up if yes.' Pause. Then cold-call 3 students: 'Tell Kamau — in one sentence — what you now know that he did not know before.' Write their three sentences on the board. Say: 'This is mathematics in the service of real Kenyan life. Kamau may not know what happens tomorrow — but he now has the tools to plan wisely.' Close the lesson.	The ritual closing — teacher re-reads the phenomenon, class answers in unison — provides narrative closure to the storyline and reinforces that the mathematics was always in service of a real human problem.	serve as holistic formative evidence of unit mastery and can be used to select exemplars for the class portfolio. Any learner whose model omits Nodes 4 or 5 (addition/multiplication laws) is flagged for targeted revision before the summative assessment.

D. TEACHER REFLECTION

1. During the game design activity, did all groups correctly identify whether their game was fair by computing theoretical probabilities — or did groups rely on intuition? If intuition dominated, how will I reinforce the use of formal calculation in fairness evaluations going forward?

2. When learners compared the experimental probability (0.4) to the naive independent-seats theoretical model (≈0.021), how many students were genuinely surprised by the mismatch? Did they articulate a meaningful real-world reason for the discrepancy (e.g., clustering due to market days or school groups), or did they remain confused? What follow-up is needed?

3. Were the Final Explanation Sentences precise enough to satisfy all three criteria (acknowledges uncertainty, cites a specific tool, connects to practical action)? What

proportion of the class achieved all three — and what does this tell me about readiness for the summative assessment?

4. Did the DQB review reveal any questions that I thought were answered but that students still marked as 'partially answered' (yellow dot)? If so, which concepts do these questions point to, and how will I address them in the summative task design?

5. Looking at the completed final models side by side with students' Lesson 1 notes: is there evidence of genuine conceptual growth from 'probability is just guessing' to 'probability is a mathematical framework for quantifying uncertainty and guiding decisions'? Which students show the most growth, and which still need support?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that Kamau's 20-trip dataset yields an experimental probability of 8/20 = 0.4 for a fully booked trip, and that this value is dramatically different from what a naive independent-seats model predicts (≈0.021) — revealing that real passenger boarding is clustered, not independent. Students also observed, through peer game evaluation, that a game is only mathematically fair when every player's theoretical probability of winning is equal, and that this must be proven with a calculation, not assumed.
What did I learn?	Learners consolidated the complete probability framework: experimental probability estimates likelihood from data and converges toward theoretical probability as trials increase (Law of Large Numbers); theoretical probability assumes equally likely outcomes and a known sample space; the Addition Law handles mutually exclusive events; the Multiplication Law and tree diagrams handle independent sequential events; and when a theoretical model's prediction contradicts experimental data, the model's assumptions (here, independence) must be questioned. A Probability Report translates all of this into actionable advice for a decision-maker like Kamau.
How does this explain the phenomenon?	The mathematics of probability explains why Kamau cannot predict any single trip with certainty — each trip is a random event influenced by factors (school terms, market days, social groups) that create clustering rather than independence — yet it empowers him to plan wisely: with P(full trip) ≈ 0.4, he can expect roughly 2 fully booked trips out of every 5 morning departures, budget for a second vehicle on approximately 8 days per school-term month, and communicate confidently to his employer that uncertainty is not ignorance but a quantifiable, manageable feature of his route.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.